

**5.0.1 UGC NET CSE | December 2013 | Part 2 | Question: 43**

Which of the following is widely used inside the telephone system for long-haul data traffic?

- A. ISDN
- B. ATM
- C. Frame Relay
- D. ISTN

ugcnetcse-dec2013-paper2 computer-networks

Answer key

5.0.2 UGC NET CSE | December 2015 | Part 3 | Question: 11

A network with bandwidth of 10 Mbps can pass only an average of 15,000 frames per minute with each frame carrying an average of 8,000 bits. What is the throughput of this network ?

- A. 2 Mbps
- B. 60 Mbps
- C. 120 Mbps
- D. 10 Mbps

computer-networks ugcnetcse-dec2015-paper3

Answer key

5.0.3 UGC NET CSE | June 2014 | Part 2 | Question: 04

The bit rate of a signal is 3000 bps. If each signal unit carries 6 bits, the baud rate of the signal is _____.

- A. 500 baud/sec
- B. 1000 baud/sec
- C. 3000 baud/sec
- D. 18000 baud/sec.

ugcnetcse-june2014-paper2 computer-networks serial-communication

Answer key

5.0.4 UGC NET CSE | June 2012 | Part 2 | Question: 5

Networks that use different technologies can be connected by using

- A. Packets
- B. Switches
- C. Bridges
- D. Routers

ugcnetcse-june2012-paper2 computer-networks

Answer key

5.0.5 UGC NET CSE | June 2012 | Part 2 | Question: 14

Which of the following TCP/IP Internet protocol does diskless machine uses to obtain its IP address from a server?

- A. RAP
- B. RIP
- C. ARP
- D. X.25

ugcnetcse-june2012-paper2 computer-networks

Answer key

5.0.6 UGC NET CSE | June 2012 | Part 2 | Question: 34

Checksum used along with each packet computes the sum of the data, where data is treated as a sequence of

- A. Integer
- B. Character
- C. Real numbers
- D. Bits

computer-networks ugcnetcse-june2012-paper2

Answer key

5.0.7 UGC NET CSE | June 2012 | Part 3 | Question: 54



What is the size (in terms of bits) of Header length field in IPV4 header?

- A. 2 B. 4 C. 8 D. 16

ugcnetcse-june2012-paper3 computer-networks

Answer key

5.0.8 UGC NET CSE | July 2016 | Part 3 | Question: 25



If link transmits 4000 frames per second and each slot has 8 bits, the transmission rate of circuits of this TDM is _____

- A. 64 Kbps B. 32 Mbps C. 32 Kbps D. 64 Mbps

ugcnetcse-july2016-paper3 computer-networks

Answer key

5.0.9 UGC NET CSE | June 2014 | Part 3 | Question: 67



Ten signals, each requiring 3000 Hz, are multiplexed on to a single channel using FDM. How much minimum bandwidth is required for the multiplexed channel? Assume that the guard bands are 300 Hz wide.

- A. 30,000 B. 32,700
C. 33,000 D. None of the above

ugcnetjune2014iii computer-networks

Answer key

5.0.10 UGC NET CSE | June 2013 | Part 2 | Question: 17



In classful addressing, the IP address 190.255.254.254 belongs to

- A. Class A B. Class B C. Class C D. Class D

ugcnetcse-june2013-paper2 computer-networks easy

Answer key

5.0.11 UGC NET CSE | July 2016 | Part 3 | Question: 27



A Pure ALOHA Network transmits 200 bit frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 500 frames per sec, then the throughput of the system is

- A. 0.384 B. 0.184 C. 0.286 D. 0.586

ugcnetcse-july2016-paper3 computer-networks pure-aloha

Answer key

5.0.12 UGC NET CSE | September 2013 | Part 2 | Question: 5



The _____ is a set of standards that defines how a dynamic web document should be written, how input data should be supplied to the program, and how the output result should be used.

- A. Hyper Text Markup Language B. File Transfer Protocol
C. Hyper Text Transfer Protocol D. Common Gateway Interface

ugcnetsep2013iii computer-networks

Answer key

5.0.13 UGC NET CSE | December 2013 | Part 3 | Question: 47



The start and stop bits are used in serial communication for

- A. error detection B. error correction

C. synchronization

D. slowing down the communication

ugcnetcse-dec2013-paper3 computer-networks serial-communication

Answer key 

5.0.14 UGC NET CSE | Junet 2015 | Part 3 | Question: 66



In a binary Hamming Code the number of check digits is r then number of message digits is equal to

A. $2^r - 1$

B. $2^r - r - 1$

C. $2^r - r + 1$

D. $2^r + r - 1$

ugcnetcse-june2015-paper3 computer-networks hamming-code

Answer key 

5.0.15 UGC NET CSE | December 2015 | Part 2 | Question: 22



Which of the following fields in IPv4 datagram is not related to fragmentation?

A. Type of service

B. Fragment offset

C. Flags

D. Identification

ugcnetcse-dec2015-paper2 computer-networks

Answer key 

5.0.16 UGC NET CSE | December 2015 | Part 2 | Question: 23



Four channels are multiplexed using TDM. If each channel sends 100 bytes/second and we multiplex 1 byte per channel, then the bit rate for the link is

A. 400 bps

B. 800 bps

C. 1600 bps

D. 3200 bps

ugcnetcse-dec2015-paper2 computer-networks

Answer key 

5.0.17 UGC NET CSE | December 2015 | Part 3 | Question: 15



A network device is sending out the data at the rate of 2000 bps. How long does it take to send a file of 1,00,000 characters?

A. 50

B. 200

C. 400

D. 800

ugcnetcse-dec2015-paper3 computer-networks

Answer key 

5.0.18 UGC NET CSE | December 2015 | Part 3 | Question: 68



Which of the following steps is/are not required for analog to digital conversion?

i. Sensing

ii. Conversion

iii. Amplification

iv. Conditioning

v. Quantization

A. a and b

B. c and d

C. a, b and e

D. None of the above

ugcnetcse-dec2015-paper3 computer-networks

Answer key 

5.0.19 UGC NET CSE | July 2016 | Part 2 | Question: 26



A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. What is the bit rate?

A. 100 Kbps

B. 200 Kbps

C. 400 Kbps

D. 1000 Kbps

Answer key

5.0.20 UGC NET CSE | July 2016 | Part 3 | Question: 26



Given the following statements:

- i. Frequency Division Multiplexing is a technique that can be applied when a bandwidth of a link is greater than combined bandwidth of signals to be transmitted
- ii. Wavelength Division Multiplexing (WDM) is an analogue multiplexing technique to combine optical signals
- iii. WDM is a digital multiplexing technique
- iv. TDM is a digital multiplexing technique

Which of the following is correct?

- A. i, ii, iii and iv are true
- B. i, ii, iii and iv are false
- C. i, ii, and iv are false; iii is true
- D. i, ii, and iv are true; iii is false

ugcnetcse-july2016-paper3 computer-networks

Answer key

5.0.21 UGC NET CSE | July 2016 | Part 3 | Question: 28



Match the following :

- | | |
|---------------------------|---|
| (a) Line Coding | (i) A technique to change analog signal to digital data |
| (b) Block Coding | (ii) Provides synchronization without increasing number of bits |
| (c) Scrambling | (iii) Process of converting digital data to digital signal |
| (d) Pulse code Modulation | (iv) Provides redundancy to ensure synchronization and inherits error detection |

Codes :

- A. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- B. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- C. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
- D. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

ugcnetcse-july2016-paper3 computer-networks

Answer key

5.0.22 UGC NET CSE | December 2011 | Part 2 | Question: 39



All of the following are examples of real security and privacy risks except

- A. Hackers
- B. Spam
- C. Viruses
- D. Identity theft

computer-networks

Answer key

5.0.23 UGC NET CSE | December 2011 | Part 2 | Question: 40



Identify the incorrect statement :

- A. The ATM adoption layer is not service dependent.
- B. Logical connections in ATM are referred to as virtual channel connections.
- C. ATM is streamlined protocol with minimal error and flow control capabilities.
- D. ATM is also known as cell delays.

computer-networks

Answer key

5.0.24 UGC NET CSE | December 2011 | Part 2 | Question: 42

The number of bits required for an IP_{V_6} address is

- A. 16 B. 32 C. 64 D. 128

computer-networks

Answer key

5.0.25 UGC NET CSE | December 2010 | Part 2 | Question: 10

Encoding of data bits 0011 into 7-bit even Parity Hamming Code is

- A. 0011110 B. 0101110 C. 0010110 D. 0011100

ugcnetcse-dec2010-paper2 computer-networks hamming-code

Answer key

5.0.26 UGC NET CSE | June 2010 | Part 2 | Question: 27

What does the URL need to access documents ?

- I. Path name
- II. Host name
- III. DNS
- IV. Retrieval method
- V. Server port number

- A. I, II, III B. I, III, V C. I, II, IV D. III, IV, V

ugcnetcse-june2010-paper2 computer-networks

Answer key

5.0.27 UGC NET CSE | June 2010 | Part 2 | Question: 46

The cost of the network is usually determined by

- A. Time complexity B. Switching complexity
C. circuit complexity D. none of these

ugcnetcse-june2010-paper2 computer-networks

Answer key

5.0.28 UGC NET CSE | January 2017 | Part 3 | Question: 64

Let C be a binary linear code with minimum distance $2t + 1$ then it can correct upto ____ bits of error

- A. $t + 1$ B. t C. $t - 2$ D. $t/2$

ugcnetcse-jan2017-paper3 computer-networks hamming-code

Answer key

5.0.29 UGC NET CSE | December 2016 | Part 3 | Question: 73

Which of the following neural networks uses supervised learning?

- i. Multilayer perceptron
- ii. Self organizing feature map
- iii. Hopfield network

- A. i only B. ii only C. i and ii only D. i and iii only

computer-networks

Answer key

5.0.30 UGC NET CSE | December 2009 | Part 2 | Question: 30



Match the following :

(i) Ethernet	(a) Deterministic
(ii) Token Ring N	(b) Utilize the full wire speed
(iii) Cut-through switch	(c) Prevent looping
(iv) Spanning tree	(d) Checking Valid address Codes

- A. a-ii, b-iii, c-iv, d-i
C. a-ii, b-iv, c-iii, d-i

- B. a-i, b-iii, c-iv, d-ii
D. a-iv, b-iii, c-ii, d-i

ugcnetcse-dec2009-paper2 computer-networks

Answer key

5.0.31 UGC NET CSE | June 2009 | Part 2 | Question: 06



The throughput of pure ALOHA is given by :

- A. $S=G$
C. $S=Ge^{2G}$

- B. $S=e^{2G}$
D. Ge^{2G}

ugcnetcse-june2009-paper2 computer-networks

5.0.32 UGC NET CSE | June 2009 | Part 2 | Question: 07



The Fiber Distributed Data Interface uses :

- A. single mode fibers and LEDs
B. multimode fibers and LEDs
C. single mode fibers and ILDs
D. multimode fibers and ILDs

ugcnetcse-june2009-paper2

Answer key

5.0.33 UGC NET CSE | June 2009 | Part 2 | Question: 08



To employ multi-access in GSM, users are given different :

1. time slots
2. bandpass filters
3. handsets
4. frequency bands

ugcnetcse-june2009-paper2

Answer key

5.0.34 UGC NET CSE | June 2009 | Part 2 | Question: 29



X.25 protocol consists of :

- A. Physical and Frame levels
B. Frame and Packet levels
C. Physical, Frame and Packet levels
D. None of the above

ugcnetcse-june2009-paper2

Answer key 

5.0.35 UGC NET CSE | June 2009 | Part 2 | Question: 30



GSM/CDMA systems :

- A. are limited to very low speed data
- B. require no local loop wires
- C. are predominantly used for voice
- D. all of the above

ugcnetcse-june2009-paper2

Answer key 

5.0.36 UGC NET CSE | June 2009 | Part 2 | Question: 31



Usually information security in a network is achieved by :

- A. Layering
- B. Cryptography
- C. Grade of service
- D. None of the above

ugcnetcse-june2009-paper2

5.0.37 UGC NET CSE | June 2009 | Part 2 | Question: 41



Which one of the following ISO standard is used for software process ?

- A. *ISO9000*
- B. *ISO9001*
- C. *ISO9003*
- D. *ISO9000 – 3*

ugcnetcse-june2009-paper2

Answer key 

5.0.38 UGC NET CSE | June 2009 | Part 2 | Question: 45



When compared with analog cellular systems, an advantage of digital *TDMA* cellular system is that :

- A. it is less complicated
- B. it requires less of computer memory
- C. it conserves spectrum bandwidth
- D. it costs less

ugcnetcse-june2009-paper2

Answer key 

5.0.39 UGC NET CSE | June 2009 | Part 2 | Question: 46



E-commerce includes :

- A. *B2C*
- B. *B2B*
- C. *C2C*
- D. All of the above

ugcnetcse-june2009-paper2

Answer key 

5.0.40 UGC NET CSE | June 2009 | Part 2 | Question: 47



A clustering technique that permits a convenient graphical display is :

- A. partition based clustering

- B. probabilistic model based clustering
- C. hierarchical clustering
- D. agglomerative clustering

ugcnetcse-june2009-paper2

Answer key 

5.0.41 UGC NET CSE | November 2017 | Part 3 | Question: 28



Match the following :

List-I

- (a) Data Link Layer
- (b) Network Layer
- (c) Transport Layer
- (d) Presentation Layer

List-II

- (i) Encryption
- (ii) Connection control
- (iii) Routing
- (iv) Framing

Code :

- A. (a)-(iv); (b)-(iii); (c)-(i); (d)-(ii)
- B. (a)-(iii); (b)-(iv); (c)-(ii); (d)-(i)
- C. (a)-(iv); (b)-(ii); (c)-(iii); (d)-(i)
- D. (a)-(iv); (b)-(iii); (c)-(ii); (d)-(i)

ugcnetcse-nov2017-paper3 computer-networks

Answer key 

5.0.42 UGC NET 2013



In the _____ method , after the station finds the line idle , it sends or refrains from sending based on probability

- a)None-Persistent
- b)One -Persistent
- c) P-Persistent
- d)Non-Persistent

Answer key 

5.0.43 UGC NET CSE | July 2018 | Part 2 | Question: 44



Which of the following statements are true?

- i. Advanced Mobile Phone System (AMPS) is a second generation cellular phone system
- ii. IS-95 is a second generation cellular phone system based on CDMA and DSSS
- iii. The Third generation cellular phone system will provide universal personnel communication

- A. i and ii only
- B. ii and iii only
- C. i, ii, and iii
- D. i and iii only

ugcnetcse-july2018-paper2 computer-networks

Answer key 

5.0.44 UGC NET CSE | December 2018 | Part 2 | Question: 89



Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- A. HTTP GET request, DNS query, TCP SYN
C. TCP SYN, DNS query, HTTP GET request

- B. DNS query, HTTP GET request, TCP SYN
D. DNS query, TCP SYN, HTTP GET request

ugcnetcse-dec2018-paper2 computer-networks

Answer key 

5.0.45 UGC NET CSE | December 2018 | Part 2 | Question: 86



A host is connected to a department network which is part of a university network. The university network, in turn, is part of the Internet. The largest network, in which the Ethernet address of the host is unique, is

- A. the Internet
C. the department network
B. the university network
D. the subnet to which the host belongs

ugcnetcse-dec2018-paper2 computer-networks

Answer key 

5.0.46 UGC NET CSE | December 2018 | Part 2 | Question: 63



Consider *ISO – OSI* network architecture reference model. Session layer of this model offers dialog control, token management and _____ as services.

- A. Synchronization
B. Asynchronization
C. Flow control
D. Errors

ugcnetcse-dec2018-paper2 computer-networks

Answer key 

5.0.47 UGC NET CSE | December 2004 | Part 2 | Question: 30



Virtual circuit is associated with a _____ service.

- A. Connectionless
C. Segmentation
B. Error-free
D. Connection-oriented

ugcnetcse-dec2004-paper2

Answer key 

5.0.48 UGC NET CSE | December 2004 | Part 2 | Question: 29



Which transport class should be used with residual-error network layer ?

- A. *TP0, TP2*
C. *TP1, TP3, TP4*
B. *TP1, TP3*
D. *TP0, TP1, TP2, TP3, TP4*

ugcnetcse-dec2004-paper2

Answer key 

5.0.49 UGC NET CSE | December 2004 | Part 2 | Question: 28



Which transport class should be used with a perfect network layer ?

- A. *TP0* and *TP2*
C. *TP0, TP1, TP3*
B. *TP1* and *TP3*
D. *TP0, TP1, TP2, TP3, TP4*

ugcnetcse-dec2004-paper2

Answer key 

5.0.50 UGC NET CSE | December 2004 | Part 2 | Question: 27



Making sure that all the data packets of a message are delivered to the destination is _____ control.

- A. Error
B. Loss
C. Sequence
D. Duplication

ugcnetcse-dec2004-paper2

Answer key 

5.0.51 UGC NET CSE | December 2004 | Part 2 | Question: 26



Error control is needed at the transport layer because of potential error occurring _____.

- A. from transmission line noise
- B. in router
- C. from out of sequency delivery
- D. from packet losses

ugcnetcse-dec2004-paper2

Answer key

5.0.52 UGC NET CSE | December 2006 | Part 2 | Question: 27



A 10 BASE-2 network is limited to :

- A. 20 bytes per data field
- B. 30 stations per segment
- C. 40 segments
- D. 50 feet of cable

ugcnetcse-dec2006-paper2

Answer key

5.0.53 UGC NET CSE | December 2006 | Part 2 | Question: 26



How many DS1 signals are transported on a DS3 signal ?

- A. 24
- B. 672
- C. 14
- D. 28

ugcnetcse-dec2006-paper2

Answer key

5.0.54 UGC NET CSE | December 2007 | Part 2 | Question: 29



For pure ALOHA, the maximum channel utilization is :

- A. 100%
- B. 50%
- C. 36%
- D. 18%

ugcnetcse-dec2007-paper2

Answer key

5.0.55 UGC NET CSE | December 2007 | Part 2 | Question: 28



The maximum data rate of binary signals on a noiseless 3 KHz channel is :

- A. 3000 bps.
- B. 6000 bps.
- C. 9000 bps.
- D. 12,000 bps.

ugcnetcse-dec2007-paper2

Answer key

5.0.56 UGC NET CSE | December 2007 | Part 2 | Question: 27



In a broadcast network, a layer that is often thin or non-existent is :

- A. network layer
- B. transport layer
- C. presentation layer
- D. application layer

ugcnetcse-dec2007-paper2

Answer key

5.0.57 UGC NET CSE | December 2007 | Part 2 | Question: 26



The best known example of a MAN is :

- A. Ethernet
- B. Cable Television
- C. FDDI
- D. IEEE 802.3

ugcnetcse-dec2007-paper2

Answer key 

5.0.58 UGC NET CSE | December 2007 | Part 2 | Question: 4



Let $e : B^m \rightarrow B^n$ is a group code. The minimum distance of 'e' is equal to :

- A. the maximum weight of a non zero code word
C. m
- B. the minimum weight of a non zero code word
D. n

ugcnetcse-dec2007-paper2

Answer key 

5.0.59 UGC NET CSE | October 2020 | Part 2 | Question: 88



Given below are two statements:

Statement *I*: The laws of nature put two fundamental limits on data rate of a channel. The H.Nyquist limit restricts the number of independent samples per second to twice the band-width in a Noiseless channel

Statement *II*: Shannon's major result about noised channel is that maximum data rate of a channel whose band width is H Hz, and whose signal-to-noise ratio is S/N is given by: Maximum number of bits/sec = $H \log_2(1 + \frac{S}{N})$

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement *I* and Statement *II* are true
B. Both Statement *I* and Statement *II* are false
C. Statement *I* is correct but Statement *II* is false
D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper2 computer-networks

Answer key 

5.0.60 UGC NET CSE | October 2020 | Part 2 | Question: 31



Consider a code with only four valid code words: 0000000000, 0000011111, 1111100000, and 1111111111. This code has distance 5. If the code word arrived is 0000000111 then the original code word must be _____

- A. 0000011111 B. 0000000000 C. 1111100000 D. 1111111111

ugcnetcse-oct2020-paper2 computer-networks hamming-code

Answer key 

5.0.61 UGC NET CSE | June 2023 | Part 1: 9



Identify the correct order of the terms A-E to complete the paragraph about MAC addresses and IP addresses given below.

A MAC address is a media access _____ address. A network device has a _____ MAC address that can help _____ the device in the network. An IP address is an Internet _____ address. An IP address can be static or _____.

- i. Protocol
ii. Control
iii. Dynamic
iv. Identify
v. E. unique

Choose the correct answer from the options given below:

- A. B, C, D, A, E B. C, E, D, A, B
C. A, B, E, D, C D. B, E, D, A, C

Answer key **5.1 Analog & Digital Transmission (1)****5.1.1 Analog & Digital Transmission: UGC NET CSE | Junet 2015 | Part 3 | Question: 15**

The process of dividing an analog signal into a string of discrete outputs, each of constant amplitude, is called:

- A. Strobing B. Amplification C. Conditioning D. Quantization

ugcnetcse-june2015-paper3 analog-&-digital-transmission computer-networks

Answer key **5.2 Analog Signals (1)****5.2.1 Analog Signals: UGC NET CSE | December 2014 | Part 2 | Question: 26**

The period of a signal is 10 ms. What is its frequency in Hertz ?

- A. 10 B. 100 C. 1000 D. 10000

ugcnetcse-dec2014-paper2 computer-networks analog-signals

Answer key **5.3 Application Layer (1)****5.3.1 Application Layer: UGC NET CSE | January 2017 | Part 2 | Question: 27**

The maximum size of the data that the application layer can pass on to the *TCP* layer below is _____.

- A. 2^{16} bytes B. 2^{16} bytes + *TCP* header length
C. 2^{16} bytes – *TCP* header length D. 2^{15} bytes

ugcnetjan2017ii computer-networks application-layer

Answer key **5.4 Asymmetric Key Cryptography (1)****5.4.1 Asymmetric Key Cryptography: UGC NET CSE | December 2014 | Part 3 | Question: 30**

Which one of the following is true for asymmetric-key cryptography ?

- A. Private key is kept by the receiver and public key is announced to the public.
B. Public key is kept by the receiver and private key is announced to the public.
C. Both private key and public key are kept by the receiver.
D. Both private key and public key are announced to the public.

ugcnetcse-dec2014-paper3 computer-networks asymmetric-key-cryptography

Answer key **5.5 Asymptotic Notation (1)****5.5.1 Asymptotic Notation: UGC NET CSE | June 2013 | Part 2 | Question: 35**

Give as good a big-O estimate as possible for the following functions:

$(n \log n + n^2)(n^3 + 2)$ and $(n! + 2^n)(n^3 + \log(n^2 + 1))$

- A. $O(n^5 + 2n^2)$ & $O(n^{3*} n!)$ B. $O(n^5)$ & $O(n^{3*} 2^n)$
C. $O(n^5)$ & $O(n^{3*} n!)$ D. $O(n^5 + 2n^2)$ & $O(n^{3*} 2^n)$

ugcnetcse-june2013-paper2 asymptotic-notation

Answer key 

5.6 Binary Search Tree (1)

5.6.1 Binary Search Tree: UGC NET CSE | December 2010 | Part 2 | Question: 25



Which of the following can be the sequence of nodes examined in binary search tree while searching for key 88

- A. 90, 40, 65, 50, 88
- B. 90, 110, 80, 85, 88
- C. 190, 60, 90, 85, 88
- D. 65, 140, 80, 70, 88

ugcnetcse-dec2010-paper2 computer-networks binary-search-tree

Answer key 

5.7 Binary Symmetric Channel (1)

5.7.1 Binary Symmetric Channel: UGC NET CSE | June 2013 | Part 3 | Question: 53



Binary symmetric channel uses

- A. Half duplex protocol
- B. Full duplex protocol
- C. Bit oriented protocol
- D. None of the above

ugcnetcse-june2013-paper3 computer-networks binary-symmetric-channel

Answer key 

5.8 Bit Oriented Protocol (1)

5.8.1 Bit Oriented Protocol: UGC NET CSE | December 2014 | Part 3 | Question: 29



_____ is a bit-oriented protocol for communication over point-to-point and multipoint links.

- A. Stop-and-wait
- B. HDLC
- C. Sliding window
- D. Go-back-N

ugcnetcse-dec2014-paper3 computer-networks bit-oriented-protocol

Answer key 

5.9 Bit Rate (1)

5.9.1 Bit Rate: UGC NET CSE | December 2014 | Part 2 | Question: 29



An analog signal has a bit rate of 6000 bps and a baud rate of 2000 baud. How many data elements are carried by each signal element

- A. 0.336 bits/baud
- B. 3 bits/baud
- C. 120,00,000 bits/baud
- D. None of the above

ugcnetcse-dec2014-paper2 computer-networks bit-rate

Answer key 

5.10 Bridges (1)

5.10.1 Bridges: UGC NET CSE | December 2008 | Part 2 | Question: 36



A conventional LAN bridge specifies only the functions of OSI :

- A. layers 1 and 2
- B. layers 1 through 3
- C. all layers
- D. none of the above

ugcnetcse-dec2008-paper2 computer-networks osi-model bridges

Answer key 

5.11

CRC Polynomial (2)

5.11.1 CRC Polynomial: UGC NET CSE | December 2013 | Part 3 | Question: 33



The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is

- A. 110010011001 B. 11001001 C. 110010011001001 D. 11001001011

ugcnetcse-dec2013-paper3 computer-networks crc-polynomial

Answer key

5.11.2 CRC Polynomial: UGC NET CSE | January 2017 | Part 3 | Question: 27



Let $G(x)$ be generator polynomial used for CRC checking. The condition that should be satisfied by $G(x)$ to correct odd numbered error bits, will be:

- A. $(1+x)$ is factor of $G(x)$ B. $(1-x)$ is factor of $G(x)$
C. $(1+x^2)$ is factor of $G(x)$ D. x is factor of $G(x)$

ugcnetcse-jan2017-paper3 computer-networks crc-polynomial

Answer key

5.12

Cache Memory (1)

5.12.1 Cache Memory: UGC NET CSE | December 2008 | Part 2 | Question: 9



Suppose it takes 100 ns to access page table and 20 ns to access associative memory. If the average access time is 28 ns , the corresponding hit rate is:

- A. 100 percent
B. 90 percent
C. 80 percent
D. 70 percent

ugcnetcse-dec2008-paper2 computer-networks cache-memory

Answer key

5.13

Checksum (1)

5.13.1 Checksum: UGC NET CSE | August 2016 | Part 3 | Question: 27



In CRC checksum method, assume that given frame for transmission is 1101011011 and the generator polynomial is $G(x) = x^4 + x + 1$. After implementing CRC encoder, the encoded word sent from sender side is _____.

- A. 11010110111110 B. 11101101011011
C. 110101111100111 D. 110101111001111

ugcnetcse-aug2016-paper3 computer-networks crc-polynomial checksum

Answer key

5.14

Classful Addressing (2)

5.14.1 Classful Addressing: UGC NET CSE | December 2012 | Part 3 | Question: 15



In classful addressing, an IP address 123.23.56.4 belongs to _____ format.

- A. A B. B C. C D. D

ugcnetcse-dec2012-paper3 computer-networks ip-addressing classful-addressing

Answer key

5.14.2 Classful Addressing: UGC NET CSE | December 2014 | Part 2 | Question: 27



In a classful addressing, first four bits in Class A IP address is

- A. 1010 B. 1100 C. 1011 D. 1110

ugcnetcse-dec2014-paper2 computer-networks classful-addressing

Answer key

5.15 Classless Addressing (1)

5.15.1 Classless Addressing: UGC NET CSE | Junet 2015 | Part 2 | Question: 30



Which of the following is/are restriction(s) in classless addressing?

- A. The number of addresses needs to be a power of 2
B. The mask needs to be included in the address to define the block
C. The starting address must be divisible by the number of addresses in the block
D. All of the above

ugcnetcse-june2015-paper2 computer-networks classless-addressing

Answer key

5.16 Communication (9)

5.16.1 Communication: UGC NET CSE | December 2013 | Part 3 | Question: 43



What is the bit rate for transmitting uncompressed 800×600 pixel colour frames with 8 bits/pixel at 40 frames/second?

- A. 2.4 Mbps B. 15.36 Mbps
C. 153.6 Mbps D. 1536 Mbps

ugcnetcse-dec2013-paper3 computer-networks communication

Answer key

5.16.2 Communication: UGC NET CSE | December 2013 | Part 3 | Question: 46



A client-server system uses a satellite network, with the satellite at a height of 40,000 kms. What is the best-case delay in response to a request? (Note that the speed of light in air is 3,00,000 km/second)

- A. 133.33 m sec B. 266.67 m sec
C. 400.00 m sec D. 533.33 m sec

ugcnetcse-dec2013-paper3 communication computer-networks

Answer key

5.16.3 Communication: UGC NET CSE | December 2013 | Part 3 | Question: 48



_____ is a type of transmission impairment in which the signal loses strength due to the resistance of the transmission medium

- A. Attenuation B. Distortion C. Noise D. Decibel

ugcnetcse-dec2013-paper3 computer-networks communication non-gatecse

Answer key

5.16.4 Communication: UGC NET CSE | June 2012 | Part 2 | Question: 19



The station to hub distance in which it is 2000 meters

- A. 100 Base-T_X B. 100 Base-F_X C. 100 Base-T₄ D. 100 Base-T₁

ugcnetcse-june2012-paper2 computer-networks communication

Answer key

5.16.5 Communication: UGC NET CSE | June 2012 | Part 2 | Question: 39



For the transmission of the signal. Bluetooth wireless technology uses

- A. time division multiplexing
- B. frequency division multiplexing
- C. time division duplex
- D. frequency division duplex

computer-networks communication ugcnetcse-june2012-paper2

Answer key

5.16.6 Communication: UGC NET CSE | June 2012 | Part 2 | Question: 45



_____ is a satellite based tracking system that enables the determination of person's position

- A. Bluetooth
- B. WAP
- C. Short Message Service
- D. Global Positioning System

computer-networks communication ugcnetcse-june2012-paper2

Answer key

5.16.7 Communication: UGC NET CSE | June 2012 | Part 3 | Question: 64



Match the following with link quality measurement and handoff initiation :

- | | |
|---|--|
| (a) Networked-Controlled Handoff (NCHO) | (i) MS connect to BS |
| (b) Mobile-Assisted Handoff (MAHO) | (ii) Process via channel the target BS |
| (c) Forward Handoff | (iii) First generation Analog Cellular System |
| (d) Hard Handoff | (iv) Second generation digital cellular system |

- A. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- B. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
- C. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- D. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

computer-networks communication ugcnetcse-june2012-paper3

Answer key

5.16.8 Communication: UGC NET CSE | June 2014 | Part 2 | Question: 01



Infrared signals can be used for short range communication in a closed area using _____ propagation.

- A. Ground
- B. Sky
- C. Line of sight
- D. Space

ugcnetcse-june2014-paper2 communication non-gatecse

Answer key

5.16.9 Communication: UGC NET CSE | June 2014 | Part 3 | Question: 06



In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Mbps ?

- A. 10 bps
- B. 100 bps
- C. 1000 bps
- D. 10000 bps

ugcnetjune2014iii computer-networks communication

Answer key

5.17.1 Computer Networks: UGC NET CSE | October 2020 | Part 2 | Question: 60



Firewall is a device that filters access to the protected network from the outside network. Firewalls can filter the packets on the basis of

- i. Source IP address
- ii. Destination IP address
- iii. TCP Source Port
- iv. UDP Source Port
- v. TCP Destination Port

Choose the correct answer from the options given below:

- A. (a), (b) and (c) only
- B. (b) and (e) only
- C. (c) and (d) only
- D. (a), (b), (c), (d) and (e) only

ugcnetcse-oct2020-paper2 computer-networks network-security

Answer key

5.18

Cryptography (7)

5.18.1 Cryptography: UGC NET CSE | August 2016 | Part 3 | Question: 30



The plain text message *BAHI* encrypted with *RSA* algorithm using $e = 3, d = 7$ and $n = 33$ and the characters of the message are encoded using the values 00 to 25 for letters A to Z. Suppose character by character encryption was implemented. Then, the Cipher Text message is _____.

- A. ABHI
- B. HAQC
- C. IHBA
- D. BHQC

ugcnetcse-aug2016-paper3 cryptography computer-networks

Answer key

5.18.2 Cryptography: UGC NET CSE | December 2013 | Part 3 | Question: 32



Find the false statement:

- A. In Modern Cryptography, symmetric key algorithms use same key both for Encryption and Decryption.
- B. The symmetric cipher DES (Data Encryption Standard) was widely used in the industry for security product.
- C. The AES (Advanced Encryption Standard) cryptosystem allows variable key lengths of size 56 bits and 124 bits.
- D. Public key algorithms use two different keys for Encryption and Decryption

ugcnetcse-dec2013-paper3 computer-networks cryptography

Answer key

5.18.3 Cryptography: UGC NET CSE | December 2014 | Part 2 | Question: 30



How many distinct stages are there in *DES* algorithm, which is parameterized by a 56-bit key ?

- A. 16
- B. 17
- C. 18
- D. 19

ugcnetcse-dec2014-paper2 computer-networks cryptography

Answer key

5.18.4 Cryptography: UGC NET CSE | July 2018 | Part 2 | Question: 45



Match the following symmetric block ciphers with corresponding block and key sizes:

List-I

- (a) DES
- (b) IDEA
- (c) BLOW FISH
- (d) AES

List-II

- (i) block size 64 and key size ranges between 32 and 448
- (ii) block size 64 and key size 64
- (iii) block size 128 and key sizes 128, 192, 256
- (iv) block size 64 and key size 128

- A. a-iv; b-ii; c-i; d-iii
- C. a-ii; b-iv; c-iii; d-i

- B. a-ii; b-iv; c-i; d-iii
- D. a-iv; b-ii; c-iii; d-i

ugcnetcse-july2018-paper2 cryptography computer-networks

Answer key 

5.18.5 Cryptography: UGC NET CSE | June 2014 | Part 3 | Question: 56



Match the following cryptographic algorithms with their design issues :

List-I**List-II**

- | | |
|----------|-------------------|
| a. DES | i. Message Digest |
| b. AES | ii. Public key |
| c. RSA | iii. 56-bit key |
| d. SHA-1 | iv. 128-bit key |

- A. a-ii, b-i, c-iv, d-iii
- C. a-iii, b-iv, c-ii, d-i

- B. a-iii, b-i, c-iv, d-ii
- D. a-iv, b-i, c-ii, d-iii

ugcnetjune2014iii computer-networks network-security cryptography

Answer key 

5.18.6 Cryptography: UGC NET CSE | October 2020 | Part 2 | Question: 32



Using 'RSA' public key cryptosystem, if $p = 3$, $q = 11$ and $d = 7$, find the value of e and encrypt the number '19'

- | | | | |
|----------|----------|---------|----------|
| A. 20,19 | B. 33,11 | C. 3,28 | D. 77,28 |
|----------|----------|---------|----------|

ugcnetcse-oct2020-paper2 computer-networks network-security cryptography

Answer key 

5.18.7 Cryptography: UGC NET CSE | October 2020 | Part 2 | Question: 54



Which among the following statement(s) is(are) true?

- i. A hash function takes a message of arbitrary length and generates a fixed length code
- ii. A hash function takes a message of fixed length and generates a code of variable length
- iii. A hash function may give same hash value for distinct messages

Choose the correct answer from the options given below:

- | | | | |
|-------------|------------------------|-----------------------|--------------|
| A. (i) only | B. (ii) and (iii) only | C. (i) and (iii) only | D. (ii) only |
|-------------|------------------------|-----------------------|--------------|

ugcnetcse-oct2020-paper2 computer-networks network-security cryptography

Answer key 

5.19

Data Communication (7)

5.19.1 Data Communication: UGC NET CSE | December 2009 | Part 2 | Question: 26



Device on one network can communicate with devices on another network via a

- (A) Hub/Switch
- (B) Utility server

(C) File server

(D) Gateway

ugcnetcse-dec2009-paper2 computer-networks data-communication

Answer key 

5.19.2 Data Communication: UGC NET CSE | December 2011 | Part 2 | Question: 35



The cellular frequency reuse factor for the cluster size N is

- A. N B. N^2 C. $\frac{1}{N}$ D. $\frac{1}{N^2}$

ugcnetcse-dec2011-paper2 computer-networks data-communication

Answer key 

5.19.3 Data Communication: UGC NET CSE | December 2011 | Part 2 | Question: 37



Handoff is the mechanism that

- A. Transfer an ongoing call from one base station to another B. Initiating a new call
C. Dropping an ongoing call D. None of above

ugcnetcse-dec2011-paper2 computer-networks data-communication

Answer key 

5.19.4 Data Communication: UGC NET CSE | December 2011 | Part 2 | Question: 48



A telephone conference call is an example of which type of communications ?

- A. Same time / same place B. Same time / different place
C. Different time / different place D. Different time / same place

ugcnetcse-dec2011-paper2 computer-networks data-communication

Answer key 

5.19.5 Data Communication: UGC NET CSE | June 2010 | Part 2 | Question: 47



A leased special high-speed connection from the local telephone carrier for business users that transmits at 1.54 mbps is known as _____ carrier.

- A. T_1 B. T_2 C. T_3 D. T_4

ugcnetcse-june2010-paper2 computer-networks data-communication

Answer key 

5.19.6 Data Communication: UGC NET CSE | June 2010 | Part 2 | Question: 48



CDMA Cell uses _____ carriers of 1.25 MHz.

- A. 9 B. 18 C. 22 D. 64

ugcnetcse-june2010-paper2 computer-networks data-communication

5.19.7 Data Communication: UGC NET CSE | Junet 2015 | Part 3 | Question: 29



Suppose a digitized voice channel is made by digitizing 8 kHz bandwidth analog voice signal. It is required to sample the signal at twice the highest frequency (two samples per hertz). What is the bit rate required, if it is assumed that each sample required 8 bits?

- A. 32 kbps B. 64 kbps C. 128 kbps D. 256 kbps

ugcnetcse-june2015-paper3 computer-networks data-communication

- A. 1 kbps B. 1.25 kbps C. 2 kbps D. 10 kbps

Answer key **5.20.6 Data Link Layer: UGC NET CSE | September 2013 | Part 2 | Question: 2**

In _____ CSMA protocol, after the station finds the line idle, it sends or refrains from sending based on the outcome of a random number generated.

- A. Non-persistent B. 0- persistent C. 1- persistent D. p- persistent

ugcnetsep2013ii computer-networks data-link-layer

Answer key **5.20.7 Data Link Layer: UGC NET CSE | September 2013 | Part 3 | Question: 7**

Suppose a file of 10,000 characters is to be sent over a line at 2400 bps. Assume that the data is sent in frames. Each frame consists of 1000 characters and an overhead of 48 bits per frame. Using synchronous transmission, the total overhead time is

- A. 0.05 second B. 0.1 second C. 0.2 second D. 2.0 seconds

ugcnetcse-sep2013-paper3 computer-networks data-link-layer

Answer key **5.21 Data Mining Algorithm (1)****5.21.1 Data Mining Algorithm: UGC NET CSE | December 2008 | Part 2 | Question: 49**

An example of a data mining algorithm which uses squared error score function is :

- A. CART algorithm
B. back propagation algorithm
C. a priori algorithm
D. vector space algorithm

ugcnetcse-dec2008-paper2 neural-network data-mining-algorithm

Answer key **5.22 Data Transmission (1)****5.22.1 Data Transmission: UGC NET CSE | December 2014 | Part 3 | Question: 26**

How many characters per second (7 bits + 1 parity) can be transmitted over a 3200 bps line if the transfer is asynchronous ? (Assuming 1 start bit and 1 stop bit)

- A. 300 B. 320 C. 360 D. 400

ugcnetcse-dec2014-paper3 computer-networks data-transmission asynchronous-transmission

Answer key **5.23 Distance Vector Routing (2)****5.23.1 Distance Vector Routing: UGC NET CSE | December 2007 | Part 2 | Question: 30**

An example of an adaptive routing algorithm is :

- A. distance vector routing B. flooding
C. selective flooding D. shortest path routing

ugcnetcse-dec2007-paper2 algorithms distance-vector-routing

Answer key 

5.23.2 Distance Vector Routing: UGC NET CSE | January 2017 | Part 2 | Question: 29



Distance vector routing algorithm is a dynamic routing algorithm. The routing tables in distance vector routing algorithm are updated _____.

- A. automatically
- B. by server
- C. by exchanging information with neighbour nodes
- D. with back up database

ugcnetjan2017ii computer-networks distance-vector-routing

Answer key

5.24

Edi Layer (1)

5.24.1 Edi Layer: UGC NET CSE | December 2008 | Part 2 | Question: 48



E-commerce involves :

- A. Electronic Data Interchange
- B. Electronic mail
- C. Electronic Bulletin Boards
- D. All of the above

ugcnetcse-dec2008-paper2 e-commerce e-technologies edi-layer

5.25

Electronic Payment Systems (1)

5.25.1 Electronic Payment Systems: UGC NET CSE | December 2015 | Part 2 | Question: 50



_____ uses electronic means to transfer funds directly from one account to another rather than by cheque or cash

- A. M-Banking
- B. E-Banking
- C. O-Banking
- D. C-Banking

ugcnetcse-dec2015-paper2 e-technologies electronic-payment-systems mobile-computing computer-networks non-gatecse

Answer key

5.26

Encoding (2)

5.26.1 Encoding: UGC NET CSE | December 2012 | Part 3 | Question: 68



An example of a dictionary-based coding technique is

- A. Run-length coding
- B. Huffman coding
- C. Predictive coding
- D. LZW coding

ugcnetcse-dec2012-paper3 computer-networks network-security encoding

Answer key

5.26.2 Encoding: UGC NET CSE | June 2013 | Part 3 | Question: 55



Given code word 1110001010 is to be transmitted with even parity check bit. The encoded word to be transmitted for this code is

- A. 11100010101
- B. 11100010100
- C. 1110001010
- D. 111000101

computer-networks ugcnetcse-june2013-paper3 encoding

Answer key

5.27

Encryption Decryption (2)

5.27.1 Encryption Decryption: UGC NET CSE | July 2018 | Part 2 | Question: 50



Encrypt the Message "HELLO MY DEARZ" using Transposition Cipher with

Key { Plain Text 2413
Cipher Text 1234

A. HLLEO YM AEDRZ
C. ELHL MDOY AZER

B. EHOLL ZYM RAED
D. ELHL DOMY ZAER

ugcnetcse-july2018-paper2 cryptography encryption-decryption computer-networks

Answer key 

5.27.2 Encryption Decryption: UGC NET CSE | June 2019 | Part 2 | Question: 86



In the TCP/IP model, encryption and decryption are functions of ____ layer.

A. data link B. network C. transport D. application

ugcnetcse-june2019-paper2 encryption-decryption

Answer key 

5.28 Entropy (1)

5.28.1 Entropy: UGC NET CSE | December 2011 | Part 2 | Question: 9



The amount of uncertainty in a system of symbol is called

A. Bandwidth B. Entropy C. Loss D. Quantum

ugcnetcse-dec2011-paper2 computer-networks entropy

Answer key 

5.29 Error Correction (5)

5.29.1 Error Correction: UGC NET CSE | December 2009 | Part 2 | Question: 03



If in an error detection and correction code a message M : “You are good students” is stored as M' : Youare areyou aregood goodare goodstudents studentsgood. What is the space required to store M' in general ? (assume that ‘n’ is the length of M)

A. 2n B. 3n C. 4n D. less than 4n

ugcnetcse-dec2009-paper2 computer-networks error-correction

Answer key 

5.29.2 Error Correction: UGC NET CSE | December 2011 | Part 2 | Question: 12



Start and stop bits are used in serial communications for

A. Error detection B. Error correction
C. Synchronization D. Slowing down the communication

ugcnetcse-dec2011-paper2 computer-networks error-correction

Answer key 

5.29.3 Error Correction: UGC NET CSE | July 2018 | Part 2 | Question: 49



To guarantee correction of upto t errors, the minimum Hamming distance d_{min} in a block code must be _____

A. $t+1$ B. $t-2$ C. $2t-1$ D. $2t+1$

ugcnetcse-july2018-paper2 computer-networks error-correction error-detection

Answer key 

5.29.4 Error Correction: UGC NET CSE | June 2009 | Part 2 | Question: 02



In order that a code is error correcting, the minimum Hamming distance should be :

1. t
2. $2t - 1$
3. $2t$
4. $2t + 1$

ugcnetcse-june2009-paper2 error-correction

Answer key 

5.29.5 Error Correction: UGC NET CSE | September 2013 | Part 3 | Question: 56



If the data unit is 11111 and the divisor is 1010. In CRC method, what is the dividend at the transmission before division?

- A. 1111110000 B. 1111111010 C. 111111000 D. 111111

ugcnetcse-sep2013-paper3 computer-networks error-correction crc-polynomial

Answer key 

5.30 Error Detection Correction (1)

5.30.1 Error Detection Correction: UGC NET CSE | June 2014 | Part 3 | Question: 57



Consider a code with five valid code words of length ten :
0000000000, 0000011111, 1111100000, 1110000011, 1111111111
Hamming distance of the code is

- A. 5 B. 10 C. 8 D. 9

ugcnetjune2014iii computer-networks error-detection-correction hamming-distance

Answer key 

5.31 Ethernet (5)

5.31.1 Ethernet: UGC NET CSE | June 2013 | Part 3 | Question: 17



The baud rate of a signal is 600 baud/second. If each signal unit carries 6 bits, then the bit rate of a signal is _____

- A. 3600 B. 100 C. 6/600 D. None of the above

ugcnetcse-june2013-paper3 computer-networks ethernet

Answer key 

5.31.2 Ethernet: UGC NET CSE | June 2013 | Part 3 | Question: 22



What is the baud rate of standard 10 Mbps Ethernet?

- A. 10 megabaud B. 20 megabaud C. 30 megabaud D. 40 megabaud

ugcnetcse-june2013-paper3 computer-networks ethernet

Answer key 

5.31.3 Ethernet: UGC NET CSE | June 2014 | Part 2 | Question: 03



The minimum frame length for 10 Mbps Ethernet is _____ bytes and maximum is _____ bytes.

- A. 64 & 128 B. 128 & 1518
C. 1518 & 3036 D. 64 & 1518

ugcnetcse-june2014-paper2 computer-networks ethernet

Answer key 

5.31.4 Ethernet: UGC NET CSE | September 2013 | Part 2 | Question: 1



A file is downloaded in a home computer using 56 kbps MODEM connected to an Internet Service Provider. If the download of the file completes in 2 minutes, what is the maximum size of the data downloaded?

- A. 112 Mbits B. 6.72 Mbits C. 67.20 Mbits D. 672 Mbits

ugcnetsep2013ii computer-networks ethernet

Answer key

5.31.5 Ethernet: UGC NET CSE | September 2013 | Part 2 | Question: 4



What is the maximum length of CAT-5 UTP cable in Fast Ethernet network?

- A. 100 meters B. 200 meters C. 1000 meters D. 2000 meters

ugcnetsep2013ii computer-networks ethernet

Answer key

5.32 Firewall (1)

5.32.1 Firewall: UGC NET CSE | September 2013 | Part 3 | Question: 58



A _____ can forward or block packets based on the information in the network layer and transport layer header

- A. Proxy firewall B. Firewall
C. Packet filter firewall D. Message digest firewall

ugcnetcse-sep2013-paper3 computer-networks firewall

Answer key

5.33 Hamming Distance (1)

5.33.1 Hamming Distance: UGC NET CSE | June 2013 | Part 3 | Question: 54



Hamming distance between 100101000110 and 110111101101 is

- A. 3 B. 4 C. 5 D. 6

ugcnetcse-june2013-paper3 computer-networks hamming-distance

Answer key

5.34 Huffman Code (1)

5.34.1 Huffman Code: UGC NET CSE | June 2013 | Part 3 | Question: 14



A data file of 1,00,000 characters contains only the characters g-l, with the frequencies as indicated in table:

	g	h	i	j	k	l
Frequency in thousand	45	13	12	16	9	5

using the variable-length code by Huffman codes, the file can be encoded with

- A. 2,52,000 bits B. 2,64,000 bits
C. 2,46,000 bits D. 2,24,000 bits

ugcnetcse-june2013-paper3 computer-networks huffman-code data-link-layer

Answer key

5.35 IP Addressing (13)

5.35.1 IP Addressing: UGC NET CSE | December 2006 | Part 2 | Question: 28



The network 198.78.41.0 is a :

- A. Class A Network
- B. Class B Network
- C. Class C Network
- D. Class D Network

ugcnetcse-dec2006-paper2 computer-networks ip-addressing

Answer key

5.35.2 IP Addressing: UGC NET CSE | December 2006 | Part 2 | Question: 29



The subnet mask 255.255.255.192

- A. Extends the network portion to 16 bits
- B. Extends the network portion to 26 bits
- C. Extends the network portion to 36 bits
- D. Has no effect on the network portion of an IP address

ugcnetcse-dec2006-paper2 computer-networks ip-addressing subnetting

Answer key

5.35.3 IP Addressing: UGC NET CSE | December 2008 | Part 2 | Question: 20



IP address in B class is given by:

- A. 125.123.123.2
- B. 191.023.21.54
- C. 192.128.32.56
- D. 10.14.12.34

ugcnetcse-dec2008-paper2 computer-networks ip-addressing

Answer key

5.35.4 IP Addressing: UGC NET CSE | December 2013 | Part 3 | Question: 44



In IPV 4, the IP address 200.200.200.200 belongs to

- A. Class A
- B. Class B
- C. Class C
- D. Class D

ugcnetcse-dec2013-paper3 computer-networks ip-addressing

Answer key

5.35.5 IP Addressing: UGC NET CSE | July 2018 | Part 2 | Question: 43



The dotted-decimal notation of the following IPV4 address in binary notation is ____

10000001 00001011 00001011 11101111

- A. 111.56.45.239
- B. 129.11.10.238
- C. 129.11.11.239
- D. 111.56.11.239

ugcnetcse-july2018-paper2 computer-networks ip-addressing

Answer key

5.35.6 IP Addressing: UGC NET CSE | June 2005 | Part 2 | Question: 30



What is the correct subnet mask to use for a class- *B* address to support 30 Networks and also have the most hosts possible ?

- A. 255.255.255.0
- B. 255.255.192.0
- C. 255.255.240.0
- D. 255.255.248.0

ugcnetcse-june2005-paper2 computer-networks ip-addressing subnetting

Answer key

5.35.7 IP Addressing: UGC NET CSE | June 2012 | Part 3 | Question: 29



Which of the following IP address class is a multicast address?

- A. Class A B. Class B C. Class C D. Class D

ugcnetcse-june2012-paper3 computer-networks ip-addressing

Answer key

5.35.8 IP Addressing: UGC NET CSE | June 2014 | Part 3 | Question: 71



A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle ?

- A. 1024 B. 2048 C. 4096 D. 8192

ugcnetjune2014iii computer-networks ip-addressing

Answer key

5.35.9 IP Addressing: UGC NET CSE | June 2019 | Part 2 | Question: 88



What percentage (%) of the IPv4, IP address space do all class C addresses consume?

- A. 12.5% B. 25% C. 37.5% D. 50%

ugcnetcse-june2019-paper2 ip-addressing

Answer key

5.35.10 IP Addressing: UGC NET CSE | Junet 2015 | Part 2 | Question: 29



What is the size of the 'total length' field in *IPv4* diagram?

- A. 4 bits B. 8 bits C. 16 bits D. 32 bits

ugcnetcse-june2015-paper2 computer-networks ip-addressing

Answer key

5.35.11 IP Addressing: UGC NET CSE | November 2017 | Part 3 | Question: 29



The address of a class *B* host is to be split into subnets with a 6– bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet?

- A. 62 subnets and 1022 hosts B. 64 subnets and 1024 hosts
C. 62 subnets and 254 hosts D. 64 subnets and 256 hosts

ugcnetcse-nov2017-paper3 ip-addressing computer-networks

Answer key

5.35.12 IP Addressing: UGC NET CSE | September 2013 | Part 3 | Question: 12



Given the IP address 201.14.78.65 and the subset mask 255.255.255.224. What is the subset address?

- A. 201.14.78.32 B. 201.14.78.64 C. 201.14.78.65 D. 201.14.78.224

ugcnetcse-sep2013-paper3 computer-networks ip-addressing

Answer key

5.35.13 IP Addressing: UGC NET CSE | September 2013 | Part 3 | Question: 8



Which of the following is the size of Network bits and Host bits of Class A of IP address?

- A. Network bits 7, Host bits 24 B. Network bits 14, Host bits 16
C. Network bits 15, Host bits 31 D. Network bits 16, Host bits 16

ugcnetcse-sep2013-paper3 computer-networks ip-addressing

Answer key 

5.36

IP Packet (2)

5.36.1 IP Packet: UGC NET CSE | December 2012 | Part 2 | Question: 9



If a packet arrive with M-bit value is '1' and a fragmentation offset value '0', then it is ____ fragement.

- A. First B. Middle C. Last D. All of the above

ugcnetcse-dec2012-paper2 computer-networks ip-packet

Answer key 

5.36.2 IP Packet: UGC NET CSE | June 2014 | Part 3 | Question: 69



Which of the following is used in the options field of *IPv4*?

- A. Strict source routing B. Loose source routing
C. Time stamp D. All of the above

ugcnetjune2014iii computer-networks ip-packet routing

Answer key 

5.37

Internal Memory (1)

5.37.1 Internal Memory: UGC NET CSE | Junet 2015 | Part 3 | Question: 1



For the 8 bit word 00111001, the check bits stored with it would be 0111. Suppose when the word is read from memory, the check bits are calculated to be 1101. What is the data word that was read from memory?

- A. 10011001 B. 00011001 C. 00111000 D. 11000110

ugcnetcse-june2015-paper3 co-and-architecture internal-memory

Answer key 

5.38

Internet Security (1)

5.38.1 Internet Security: UGC NET CSE | July 2018 | Part 2 | Question: 47



In Challenge-Response authentication the claimant ____

- A. Proves that she knows the secret without revealing it B. Proves that she does not know the secret
C. Reveals the secret D. Gives a challenge

ugcnetcse-july2018-paper2 computer-networks internet-security

Answer key 

5.39

Internet Stack Layers (1)

5.39.1 Internet Stack Layers: UGC NET CSE | June 2019 | Part 2 | Question: 87



Which of the following statements is/are true with regard to various layers in the Internet stack?

P: At the link layer, a packet of transmitted information is called a frame.

Q: At the network layer, a packet of transmitted information is called a segment.

- A. P only B. Q only C. P and Q D. Neither P nor Q

ugcnetcse-june2019-paper2 internet-stack-layers

Answer key 

5.40

Ipv6 (1)

5.40.1 Ipv6: UGC NET CSE | November 2017 | Part 3 | Question: 30



Which of the following statements are true?

- i. The fragmentation fields in the base header section of IPv4 have moved to the fragmentation extension header in IPv6.
- ii. The authentication extension header is new in IPv6.
- iii. The record route option is not implemented in IPv6.

- A. (i) and (ii) Only
- C. (i) and (iii) Only

- B. (ii) and (iii) Only
- D. (i), (ii) and (iii)

ugcnetcse-nov2017-paper3 computer-networks ip-addressing ipv6

Answer key

5.41

LAN Technologies (2)

5.41.1 LAN Technologies: UGC NET CSE | December 2010 | Part 2 | Question: 27



The baud rate is

- A. Always equal to the bit transfer rate
- B. Equal to twice the bandwidth of an ideal channel
- C. Not equal to the signalling rate
- D. Equal to half of the bandwidth of an ideal channel

ugcnetcse-dec2010-paper2 computer-networks lan-technologies

Answer key

5.41.2 LAN Technologies: UGC NET CSE | January 2017 | Part 3 | Question: 25



Suppose there are N stations in a slotted LAN. Each station attempts to transmit with a probability P in each time slot. The probability that only one station transmits in a given slot is _____

- A. $nP(1 - P)^{n-1}$
- B. nP
- C. $P(1 - P)^{n-1}$
- D. $n^P(1 - P)^{n-1}$

ugcnetcse-jan2017-paper3 computer-networks lan-technologies

Answer key

5.42

Link State Routing (1)

5.42.1 Link State Routing: UGC NET CSE | January 2017 | Part 2 | Question: 30



In link state routing algorithm after construction of link state packets, new routes are computed using :

- A. DES algorithm
- B. Dijkstra's algorithm
- C. RSA algorithm
- D. Packets

ugcnetjan2017ii computer-networks link-state-routing

Answer key

5.43

Loopback Address (1)

5.43.1 Loopback Address: UGC NET CSE | June 2019 | Part 2 | Question: 85



Consider the following two statements with respect to IPv4 in computer networking:

P: The loopback (IP) address is a member of class B network.

Q: The loopback (IP) address is used to send a packet from host to itself.

What can you say about the statements P and Q?

- A. P – True; Q – False
- B. P – False; Q – True
- C. P – True; Q – True
- D. P – False; Q – False

Answer key 

5.44

Mesh topology (3)

5.44.1 Mesh topology: UGC NET CSE | December 2014 | Part 3 | Question: 25



For n devices in a network, _____ number of duplex-mode links are required for a mesh topology.

- A. $n(n+1)$ B. $n(n-1)$
C. $n(n+1)/2$ D. $n(n-1)/2$

ugcnetcse-dec2014-paper3 computer-networks network-topologies mesh-topology

Answer key 

5.44.2 Mesh topology: UGC NET CSE | July 2016 | Part 2 | Question: 27



In a fully connected mesh network with 10 computers, total ____ number of cables are required and ____ number of ports are required for each device

- A. 40, 9 B. 45, 10 C. 45, 9 D. 50, 10

ugcnetcse-july2016-paper2 computer-networks mesh-topology

Answer key 

5.44.3 Mesh topology: UGC NET CSE | June 2019 | Part 2 | Question: 81



A fully connected network topology is a topology in which there is a direct link between all pairs of nodes. Given a fully connected network with n nodes, the number of direct links as a function of n can be expressed as

- A. $\frac{n(n+1)}{2}$ B. $\frac{(n+1)}{2}$ C. $\frac{n}{2}$ D. $\frac{n(n-1)}{2}$

ugcnetcse-june2019-paper2 mesh-topology

Answer key 

5.45

Mobile Communication (7)

5.45.1 Mobile Communication: UGC NET CSE | December 2012 | Part 2 | Question: 3



The GSM network is divided into the following three major systems:

- A. SS, BSS, OSS B. BSS, BSC, MSC
C. CELL, BSS, OSS D. SS, CELL, MSC

ugcnetcse-dec2012-paper2 computer-networks communication mobile-communication

Answer key 

5.45.2 Mobile Communication: UGC NET CSE | December 2012 | Part 3 | Question: 26



Which are the classifications of data used in Mobile Applications?

- A. Private data, Shared data, User data B. Public data, User data, Virtual data
C. Private data, Public data, Shared data D. Public data, Virtual data, User data

ugcnetcse-dec2012-paper3 mobile-communication computer-networks

Answer key 

5.45.3 Mobile Communication: UGC NET CSE | December 2012 | Part 3 | Question: 30



Match the following with respect to the Mobile Computing Architecture.

- a. Downlink control 1. 100 Mbps
b. Data communication 2. Residency
data rate Latency (RL)

- c. The average duration of 3. Sending data
user's stay in cell from a BS to MD
- d. FDDI bandwidth 4. 2-Mbps

- a b c d
- A 2 1 4 3
- B 3 4 2 1
- C 4 1 2 1
- D 4 3 1 2

ugcnetcse-dec2012-paper3 mobile-communication computer-networks non-gatecse

Answer key 

5.45.4 Mobile Communication: UGC NET CSE | December 2014 | Part 2 | Question: 46



Consider the following statements S_1 and S_2 :

S_1 : A hard handover is one in which the channel in the source cell is retained and used for a while in parallel with the channel in the target cell.

S_2 : A soft handover is one in which the channel in the source cell is released and only then the channel in the target cell is engaged.

- A. S_1 is true and S_2 is not true. B. S_1 is not true and S_2 is true.
- C. Both S_1 and S_2 are true. D. Both S_1 and S_2 are not true.

ugcnetcse-dec2014-paper2 mobile-communication

Answer key 

5.45.5 Mobile Communication: UGC NET CSE | December 2015 | Part 2 | Question: 24



In a typical mobile phone system with hexagonal cells, it is forbidden to reuse a frequency band in adjacent cells. If 840 frequencies are available, how many can be used in a given cell?

- A. 280 B. 210 C. 140 D. 120

ugcnetcse-dec2015-paper2 mobile-communication computer-networks non-gatecse

Answer key 

5.45.6 Mobile Communication: UGC NET CSE | June 2012 | Part 3 | Question: 60



Radio signals generally propagate according to the following mechanisms:

- A. Modulation, Amplification, Scattering
- B. Reflection, Diffraction, Scattering
- C. Amplification, Diffraction, Modulation
- D. Reflection, Amplification, Diffraction

ugcnetcse-june2012-paper3 computer-networks transmission-media mobile-communication

Answer key 

5.45.7 Mobile Communication: UGC NET CSE | June 2014 | Part 2 | Question: 26



Match the following with respect to the mobile computing technologies :

List-I

List-II

- | | |
|---------|---|
| a. GPRS | i. An Integrated digital radio standard |
| b. GSM | ii. 3G wireless/Mobile technology |
| c. UMTS | iii. Nine different schemes for modulation and error correction |
| d. EDGE | iv. An emerging wireless service that offeres a mobile data |

Codes :

- A. a-iii; b-iv; c-ii; d-i
C. a-ii; b-iii; c-iv; d-i

ugcnetcse-june2014-paper2 computer-networks mobile-communication

Answer key 

- B. a-iv; b-i; c-ii; d-iii
D. a-ii; b-i; c-iv; d-iii

5.46

Modulation (2)

5.46.1 Modulation: UGC NET CSE | November 2017 | Part 3 | Question: 26



Quadrature Amplitude Modulation means changing both

- A. Frequency and phase of the carrier
C. Amplitude and phase of the carrier
B. Frequency and Amplitude of the carrier
D. Amplitude and Wavelength of the carrier

ugcnetcse-nov2017-paper3 computer-networks modulation physical-layer

Answer key 

5.46.2 Modulation: UGC NET CSE | September 2013 | Part 3 | Question: 10



Which of the following is an bit rate of an 8-PSK signal having 2500 Hz bandwidth?

- A. 2500 bps
B. 5000 bps
C. 7500 bps
D. 20000 bps

ugcnetcse-sep2013-paper3 computer-networks modulation

Answer key 

5.47

Multimedia Protocol (1)

5.47.1 Multimedia Protocol: UGC NET CSE | December 2015 | Part 3 | Question: 14



In electronic mail, which of the following protocols allows the transfer of multimedia messages?

- A. IMAP
B. SMTP
C. POP 3
D. MIME

ugcnetcse-dec2015-paper3 computer-networks multimedia-protocol

Answer key 

5.48

Multiple Access Protocol (1)

5.48.1 Multiple Access Protocol: UGC NET CSE | August 2016 | Part 3 | Question: 28



A slotted ALOHA network transmits 200 bits frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 1000 frames per second, then the throughput of the system is _____.

- A. 0.268
B. 0.468
C. 0.368
D. 0.568

ugcnetcse-aug2016-paper3 computer-networks multiple-access-protocol transmission-media

Answer key 

5.49

Multiplexing (2)

5.49.1 Multiplexing: UGC NET CSE | December 2008 | Part 2 | Question: 35



A high performance switching and multiplexing technology that utilizes fixed length packets to carry different types of traffic is:

- A. ATM
B. ADSL
C. SONET
D. None of the above

ugcnetcse-dec2008-paper2 computer-networks multiplexing network-switching

Answer key 

5.49.2 Multiplexing: UGC NET CSE | December 2010 | Part 2 | Question: 30



The threshold effect in demodulator is

- A. Exhibited by all demodulator, when the input signal to noise ratio is low.
- B. The rapid fall on output signal to noise ratio when the input signal to noise ratio fall below a particular value.
- C. The property exhibited by all A.M. suppressed carrier coherent demodulator.
- D. The property exhibited by correlation receiver.

ugcnetcse-dec2010-paper2 computer-networks multiplexing

Answer key

5.50 Network Addressing (2)

5.50.1 Network Addressing: UGC NET CSE | December 2015 | Part 3 | Question: 10



The combination of an IP address and a port number is known as _____

- A. network number
- B. socket address
- C. subnet mask number
- D. MAC address

computer-networks network-addressing ugcnetcse-dec2015-paper3

Answer key

5.50.2 Network Addressing: UGC NET CSE | December 2015 | Part 3 | Question: 13



In a classful addressing the IP address with 0(zero) as network number:

- A. refers to the current network
- B. refers to broadcast on the local network
- C. refers to the broadcast on a distant network
- D. refers to loopback testing

computer-networks network-addressing ugcnetcse-dec2015-paper3

Answer key

5.51 Network Layer (1)

5.51.1 Network Layer: UGC NET CSE | December 2015 | Part 2 | Question: 40



Which of the following is not a congestion policy at network layer?

- A. Flow Control Policy
- B. Packet Discard Policy
- C. Packet Lifetime Management Policy
- D. Routing Algorithm

ugcnetcse-dec2015-paper2 computer-networks network-layer

Answer key

5.52 Network Layering (5)

5.52.1 Network Layering: UGC NET CSE | December 2012 | Part 2 | Question: 17



The design issue of Datalink layer in OSI Reference Model is

- A. Framing
- B. Representation of bits
- C. Synchronization of bits
- D. Connection control

ugcnetcse-dec2012-paper2 computer-networks network-layering

Answer key

5.52.2 Network Layering: UGC NET CSE | December 2013 | Part 3 | Question: 45



Which layer of OSI reference model is responsible for decomposition of messages and generation of sequence numbers to ensure correct re-composition from end to end of the network?

- A. Physical
- B. Data-link
- C. Transport
- D. Application

Answer key **5.52.3 Network Layering: UGC NET CSE | June 2012 | Part 2 | Question: 15**

Decryption and encryption of data are responsibility of which of the following layer?

- | | |
|-----------------------|--------------------|
| A. Physical layer | B. Data link layer |
| C. Presentation layer | D. Session layer |

ugcnetcse-june2012-paper2 computer-networks network-layering

Answer key **5.52.4 Network Layering: UGC NET CSE | June 2012 | Part 2 | Question: 6**

Both hosts and routers and TCP/IP protocol software. However, routers do not use protocol from all layers. The layer for which the protocol software is not needed by a router is

- | | |
|----------------------------|----------------------------------|
| A. Layer – 5 (Application) | B. Layer – 1 (Physical) |
| C. Layer – 3 (Internet) | D. Layer – 4 (Network Interface) |

ugcnetcse-june2012-paper2 computer-networks network-layering

Answer key **5.52.5 Network Layering: UGC NET CSE | June 2014 | Part 2 | Question: 05**

Match the following :

List – I**List – II**

- | | |
|----------------------|--|
| a. Physical layer | i. Allow resources to network access |
| b. Datalink layer | ii. Move packets from one destination to other |
| c. Network layer | iii. Process to process message delivery |
| d. Transport layer | iv. Transmission of bit stream |
| e. Application Layer | v. Formation of frames |

Codes :

- | | |
|--------------------------------|--------------------------------|
| A. a-iv; b-v; c-ii; d-iii; e-i | B. a-v; b-iv; c-i; d-ii; e-iii |
| C. a-i; b-iii; c-ii; d-v; e-iv | D. a-i; b-ii; c-iv; d-iii; e-v |

ugcnetcse-june2014-paper2 computer-networks network-layering

Answer key **5.53 Network Protocols (10)****5.53.1 Network Protocols: UGC NET CSE | August 2016 | Part 2 | Question: 30**

Which of the following protocols is used by email server to maintain a central repository that can be accessed from any machine ?

- | | | | |
|----------------|----------------|----------------|----------------|
| A. <i>POP3</i> | B. <i>IMAP</i> | C. <i>SMTP</i> | D. <i>DMSP</i> |
|----------------|----------------|----------------|----------------|

ugcnetcse-aug2016-paper2 computer-networks network-protocols

Answer key **5.53.2 Network Protocols: UGC NET CSE | August 2016 | Part 3 | Question: 26**

Consider the following statements :

- High speed Ethernet works on optic fiber.
 - A point to point protocol over Ethernet is a network protocol for encapsulating *PPP* frames inside Ethernet frames.
 - High speed Ethernet does not work on optic fiber.
 - A point to point protocol over Ethernet is a network protocol for encapsulating Ethernet frames inside *PPP* frames.
- Which of the following is correct ?

- A. A and B are true; C and D are false.
C. A, B, C and D are true.

- B. A and B are false; C and D are true.
D. A, B, C and D are false.

ugcnetcse-aug2016-paper3 computer-networks ethernet network-protocols

Answer key 

5.53.3 Network Protocols: UGC NET CSE | December 2010 | Part 2 | Question: 29



Which of the following file transfer protocols use TCP and establishes two virtual circuits between the local and remote server ?

- A. FTP B. $TFTP$ C. $TELNET$ D. NFS

ugcnetcse-dec2010-paper2 computer-networks network-protocols

Answer key 

5.53.4 Network Protocols: UGC NET CSE | December 2012 | Part 2 | Question: 8



The Mobile Application Protocol (MAP) typically runs on top of which protocol?

- A. SNMP (Simple Network Management Protocol) B. SMTP (Simple Mail Transfer Protocol)
C. SS7 (Signalling System 7) D. HTTP (Hyper Text Transfer Protocol)

ugcnetcse-dec2012-paper2 computer-networks network-protocols

Answer key 

5.53.5 Network Protocols: UGC NET CSE | June 2010 | Part 2 | Question: 26



The _____ field is the SNMP PDV reports an error in a response message.

- A. Error index B. Error status C. Set request D. Agent index

ugcnetcse-june2010-paper2 computer-networks network-protocols

Answer key 

5.53.6 Network Protocols: UGC NET CSE | October 2020 | Part 2 | Question: 33



Protocols in which the sender sends one frame and then waits for an acknowledgment before proceeding for next frame are called as _____

- A. Simplex protocols B. Unrestricted simplex protocols
C. Simplex stop and wait protocols D. Restricted simplex protocols

ugcnetcse-oct2020-paper2 computer-networks network-protocols

Answer key 

5.53.7 Network Protocols: UGC NET CSE | October 2020 | Part 2 | Question: 59



Post office protocol (POP) is a message access protocol which is used to extract message for client. In this regards, which of the following are true?

- i. POP has two modes, Delete mode and keep
- ii. In Delete mode, mail is deleted from mail box after each retrieval
- iii. In Delete mode, mail is deleted from mail box before each retrieval
- iv. In keep mode, mail is deleted before retrieval
- v. In keep mode, mail remains in mail box after retrieval

Choose the correct answer from the options given below:

- A. (i) and (ii) only B. (i), (iv) and (v) only
C. (i), (ii), (iii) and (iv) only D. (i), (ii) and (v) only

ugcnetcse-oct2020-paper2 computer-networks network-protocols

Answer key 

5.53.8 Network Protocols: UGC NET CSE | October 2020 | Part 2 | Question: 72

Match List I with List II

List I

- (A) Serial Line IP (SLIP)
 (B) Border Gateway Protocol (BGP)
 (C) User Data Protocol (UDP)
 (D) Simple Network Management Protocol

List II

- (I) Application layer
 (II) Transport layer
 (III) Data Link layer
 (IV) Network layer

Choose the correct answer from the options given below:

- A. A – I, B – II, C – III, D – IV
 C. A – II, B – III, C – IV, D – I

- B. A – III, B – IV, C – II, D – I
 D. A – III, B – I, C – IV, D – II

ugcnetcse-oct2020-paper2 computer-networks network-protocols

Answer key

5.53.9 Network Protocols: UGC NET CSE | September 2013 | Part 2 | Question: 42

A method to provide secure transmission of email is called

- A. TLS B. SA C. IPSec D. PGP

ugcnetsep2013ii computer-networks network-security network-protocols

Answer key

5.53.10 Network Protocols: UGC NET CSE | September 2013 | Part 3 | Question: 11

Match the following :

- (a) UDP (i) Message Transfer Protocol
 (b) OSPF (ii) Bit Oriented Protocol
 (c) SMTP (iii) Interior Gateway Routing Protocol
 (d) HDLC (iv) Connectionless Transport Protocol

Codes:

- A. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
 C. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

- B. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 D. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

ugcnetcse-sep2013-paper3 computer-networks network-protocols

Answer key

5.54**Network Security (18)****5.54.1 Network Security: UGC NET CSE | August 2016 | Part 2 | Question: 46**

An attacker sits between the sender and receiver and captures the information and retransmits to the receiver after some time without altering the information. This attack is called as _____.

- A. Denial of service attack B. Masquerade attack
 C. Simple attack D. Complex attack

ugcnetcse-aug2016-paper2 computer-networks network-security

Answer key

5.54.2 Network Security: UGC NET CSE | December 2010 | Part 2 | Question: 20

Which of the following is the process by which a user's access to physical data in the application is limited, based on his privileges?

- A. Authorization B. Authentication

C. Access Control

D. All of these

ugcnetcse-dec2010-paper2 computer-networks network-security

Answer key 

5.54.3 Network Security: UGC NET CSE | December 2012 | Part 2 | Question: 37



AES is a round cipher based on the Rijndal Algorithm that uses a 128 – bit block of data. AES has three different configurations. _____ rounds with a key size of 128-bits, _____ rounds with a key size of 192 bits and _____ rounds with a key size of 256 –bits

A. 5, 7, 15

B. 10, 12, 14

C. 5, 6, 7

D. 20, 12, 14

ugcnetcse-dec2012-paper2 computer-networks network-security

Answer key 

5.54.4 Network Security: UGC NET CSE | December 2012 | Part 3 | Question: 72



A cryptarithmic problem of the type

SEND
+ MORE

MONEY

Can be solved efficiently using

A. depth first technique

B. breadth first technique

C. constraint satisfaction technique

D. bidirectional technique

ugcnetcse-dec2012-paper3 computer-networks network-security

Answer key 

5.54.5 Network Security: UGC NET CSE | December 2012 | Part 3 | Question: 8



Using RSA algorithm, what is the value of cipher text C, if the plain text M=5 and p=3, q=11 and d=7?

A. 33

B. 5

C. 25

D. 26

ugcnetcse-dec2012-paper3 computer-networks network-security

Answer key 

5.54.6 Network Security: UGC NET CSE | December 2015 | Part 2 | Question: 25



Using p=3, q=13, d=7 and e=3 in the RSA algorithm, what is the value of ciphertext for a plain text 5?

A. 13

B. 21

C. 26

D. 33

ugcnetcse-dec2015-paper2 computer-networks network-security

Answer key 

5.54.7 Network Security: UGC NET CSE | December 2018 | Part 2 | Question: 66



Which of the following statement/s is/are **true**?

- i. Firewalls can screen traffic going into or out of an organization.
- ii. Virtual private networks can stimulate an old leased network to provide certain desirable properties.

Choose the correct answer from the code given below:

A. (i) only

B. (ii) only

C. Both (i) and (ii)

D. Neither (i) nor (ii)

ugcnetcse-dec2018-paper2 network-security

Answer key 

5.54.8 Network Security: UGC NET CSE | December 2018 | Part 2 | Question: 90

Suppose that everyone in a group of N people wants to communicate secretly with $(N - 1)$ other people using symmetric key cryptographic system. The communication between any two persons should not be decodable by the others in the group. The number of keys required in the system as a whole to satisfy the confidentiality requirement is

- A. $N(N - 1)$ B. $N(N - 1)/2$ C. $2N$ D. $(N - 1)^2$

ugcnetcse-dec2018-paper2 computer-networks network-security

Answer key

5.54.9 Network Security: UGC NET CSE | July 2016 | Part 2 | Question: 29

If there are N people in the world and are using secret key encryption and decryption for privacy purpose, then the number of secret keys required will be

- A. N B. $(N - 1)$ C. $\frac{N(N-1)}{2}$ D. $\frac{N(N+1)}{2}$

ugcnetcse-july2016-paper2 computer-networks network-security

Answer key

5.54.10 Network Security: UGC NET CSE | July 2016 | Part 2 | Question: 46

An attacker sits between customer and banker and captures the information from the customer and retransmits to the banker by altering the information. This attack is called as

- A. Masquerade attack B. Replay attack
C. Passive attack D. Denial of service attack

ugcnetcse-july2016-paper2 computer-networks network-security

Answer key

5.54.11 Network Security: UGC NET CSE | July 2016 | Part 3 | Question: 30

Encrypt the plain text message "EXTRANET" using Transposition cipher technique with the following key :

3	5	2	1	4	(Cipher text)
1	2	3	4	5	(Plain text)

Using 'Z' as bogus character.

- A. TAXERTZENZ B. EXTRANETZZ C. EZXZTRZANZET D. EXTZRANZETZ

ugcnetcse-july2016-paper3 computer-networks network-security

Answer key

5.54.12 Network Security: UGC NET CSE | June 2009 | Part 2 | Question: 48

After sending a message, the sender should not be able to, at a later date, deny having sent the message, is referred to as :

- A. Authenticity
B. Non-Repudiability
C. Auditability
D. Repudiability

ugcnetcse-june2009-paper2 network-security

Answer key

5.54.13 Network Security: UGC NET CSE | June 2010 | Part 2 | Question: 5



For your ATM debit card, you have a 4-decimal-digit personal secret code. In the absence of any clue, a brute-force attack takes time- t' to crack the code on an ATM terminal. Therefore t' is the secure-time for a customer to report in case the card is misplaced. Your Bank has decided to facilitate an increased secure-time. Out of the following, which option should provide the largest rise in the value of t' ?

- A. Instead of 4-decimal-digits, maintain the personal secret code in 4-hexadecimal-digits.
- B. Instead of 4-decimal digits, maintain a 5-decimal-digit personal secret code.
- C. Reduce the processing speed of the ATM terminals to the half of their current speed.
- D. None of the above provides any improvement.

ugcnetcse-june2010-paper2 computer-networks network-security

Answer key

5.54.14 Network Security: UGC NET CSE | June 2014 | Part 2 | Question: 30



SET, an open encryption and security specification model that is designed for protecting credit card transactions on the internet, stands for

- A. Secure Electronic Transaction
- B. Secular Enterprise for Transaction
- C. Security Electronic Transmission
- D. Secured Electronic Termination

ugcnetcse-june2014-paper2 computer-networks network-security

Answer key

5.54.15 Network Security: UGC NET CSE | June 2019 | Part 2 | Question: 89



The ability to inject packets into the Internet with a false source address is known as

- A. Man-in-the-middle attack
- B. IP phishing
- C. IP sniffing
- D. IP spoofing

ugcnetcse-june2019-paper2 network-security

Answer key

5.54.16 Network Security: UGC NET CSE | Junet 2015 | Part 3 | Question: 27



Which are two modes of IP security?

- A. Transport and certificate
- B. Transport and tunnel
- C. Certificate and tunnel
- D. Preshared and transport

ugcnetcse-june2015-paper3 computer-networks network-security

Answer key

5.54.17 Network Security: UGC NET CSE | Junet 2015 | Part 3 | Question: 28



A message "COMPUTERNETWORK" encrypted (ignore quotes) using columnar transposition cipher with a key "LAYER". The encrypted message is

- A. CTTOEWMROPNRUEK
- B. MROUEKCTTPNROEW
- C. OEWPNRCTTUEKMRO
- D. UEKPNRMROOEQCTT

ugcnetcse-june2015-paper3 computer-networks network-security

Answer key

5.54.18 Network Security: UGC NET CSE | October 2020 | Part 2 | Question: 89



Given below are two statements:

Statement *I*: In Caesar Cipher each letter of Plain text is replaced by another letter for encryption

Statement *II*: Diffie-Hellman algorithm is used for exchange of secret key

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement *I* and Statement *II* are true
- B. Both Statement *I* and Statement *II* are false
- C. Statement *I* is correct but Statement *II* is false
- D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper2 computer-networks network-security

Answer key 

5.55

Network Switching (5)

5.55.1 Network Switching: UGC NET CSE | August 2016 | Part 3 | Question: 25



Suppose transmission rate of a channel is 32 kbps. If there are '8' routes from source to destination and each packet p contains 8000 bits. Total end to end delay in sending packet P is _____.

- A. 2 sec
- B. 3 sec
- C. 4 se
- D. 1 sec

ugcnetcse-aug2016-paper3 computer-networks network-switching

Answer key 

5.55.2 Network Switching: UGC NET CSE | December 2011 | Part 2 | Question: 10



Which of the following network access standard disassembler is used for connection station to a packet switched network ?

- A. X.3
- B. X.21
- C. X.25
- D. X.75

ugcnetcse-dec2011-paper2 computer-networks network-switching

Answer key 

5.55.3 Network Switching: UGC NET CSE | January 2017 | Part 3 | Question: 28



In a packet switching network, if the message size is 48 bytes and each packet contains a header of 3 bytes. If 24 packets are required to transmit the message, the packet size is

- A. 2 bytes
- B. 1 byte
- C. 4 bytes
- D. 5 bytes

ugcnetcse-jan2017-paper3 computer-networks network-switching

Answer key 

5.55.4 Network Switching: UGC NET CSE | July 2018 | Part 2 | Question: 46



Which of the following statements are true?

- i. Three broad categories of Networks are
 - i. Circuit Switched Networks
 - ii. Packet Switched Networks
 - iii. Message Switched Networks
- ii. Circuit Switched Network resources need not be reserved during the set up phase
- iii. In packet switching there is no resource allocation for packets

- A. i and ii only
- B. ii and iii only
- C. i and iii only
- D. i, ii and iii only

ugcnetcse-july2018-paper2 computer-networks network-switching

Answer key 

5.55.5 Network Switching: UGC NET CSE | June 2012 | Part 2 | Question: 16



In which circuit switching, delivery of data is delayed because data must be stored and retrieved from RAM?

A. Space division

B. Time division

C. Virtual

D. Packet

ugcnetcse-june2012-paper2 computer-networks network-switching

Answer key 

5.56

Network topologies (4)

5.56.1 Network topologies: UGC NET CSE | December 2009 | Part 2 | Question: 29



In case of Bus/Tree topology signal balancing issue is overcome by

(A) Modulation

(B) Polling

(C) Segmentation

(D) Strong transmitter

ugcnetcse-dec2009-paper2 computer-networks network-topologies

5.56.2 Network topologies: UGC NET CSE | December 2009 | Part 2 | Question: 46



The single stage network is also called

(A) one sided network

(B) two sided network

(C) recirculating network

(D) pipeline network

ugcnetcse-dec2009-paper2 computer-networks network-topologies

Answer key 

5.56.3 Network topologies: UGC NET CSE | December 2011 | Part 2 | Question: 8



Hub is a term used with

A. A Star Networks

B. A Ring Networks

C. A Router

D. A Bridge

ugcnetcse-dec2011-paper2 computer-networks network-topologies

Answer key 

5.56.4 Network topologies: UGC NET CSE | June 2013 | Part 3 | Question: 16



In a fully connected mesh network with n devices, there are _____ physical channels to link all devices.

A. $n(n-1)/2$

B. $n(n+1)/2$

C. $2n$

D. $2n+1$

ugcnetcse-june2013-paper3 computer-networks network-topologies

Answer key 

5.57

Osi Model (11)

5.57.1 Osi Model: UGC NET CSE | August 2016 | Part 2 | Question: 29



Match the following :

List-I

- a. Session layer
- b. Application layer
- c. Presentation layer
- d. Transport Layer

List-II

- i. Virtual terminal software
- ii. Semantics of the information transmitted
- iii. Flow Control
- iv. Manage dialogue Control

Codes :

- A. a-iv, b-i, c-ii, d-iii
- C. a-iv, b-i, c-iii, d-ii

- B. a-i, b-iv, c-ii, d-iii
- D. a-iv, b-ii, c-i, d-iii

ugcnetcse-aug2016-paper2 computer-networks osi-model

Answer key **5.57.2 Osi Model: UGC NET CSE | December 2008 | Part 2 | Question: 17**

Congestion control is done by :

- A. Network layer
- B. Physical layer
- C. Presentation layer
- D. Application layer

ugcnetcse-dec2008-paper2 computer-networks osi-model

Answer key **5.57.3 Osi Model: UGC NET CSE | December 2008 | Part 2 | Question: 18**

Assertion (A): Twisted pairs are widely used as transmission medium.

Reasoning (R): Twisted pairs have adequate performance and low cost.

- A. Both (A) and (R) are true and (R) is the correct explanation for (A)
- B. Both (A) and (R) are true but (R) is not the correct explanation
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

ugcnetcse-dec2008-paper2 computer-networks osi-model

Answer key **5.57.4 Osi Model: UGC NET CSE | December 2008 | Part 2 | Question: 6**

An example of a layer that is absent in broadcast networks is :

- A. Physical layer
- B. Presentation layer
- C. Network layer
- D. Application layer

ugcnetcse-dec2008-paper2 computer-networks osi-model network-layer

Answer key **5.57.5 Osi Model: UGC NET CSE | December 2012 | Part 2 | Question: 43**

Data Encryption Techniques are particularly used for _____

- A. protecting data in Data Communication System
- B. reduce Storage Space Requirement
- C. enhances Data Integrity
- D. decreases Data Integrity

ugcnetcse-dec2012-paper2 computer-networks osi-model

Answer key 

5.57.6 Osi Model: UGC NET CSE | December 2013 | Part 2 | Question: 1



When data and acknowledgement are sent in same frame, this is called as

- A. Piggy packing
- B. Piggy backing
- C. Back packing
- D. Good packing

ugcnetcse-dec2013-paper2 computer-networks osi-model

Answer key

5.57.7 Osi Model: UGC NET CSE | December 2013 | Part 2 | Question: 2



Encryption and decryption is the responsibility of ____ Layer

- A. Physical
- B. Network
- C. Application
- D. Datalink

ugcnetcse-dec2013-paper2 computer-networks osi-model

Answer key

5.57.8 Osi Model: UGC NET CSE | June 2010 | Part 2 | Question: 28



End-to-End connectivity is provided from Last-to-Last in

- A. Network layer
- B. Session layer
- C. Transport layer
- D. Data link layer

ugcnetcse-june2010-paper2 computer-networks osi-model

Answer key

5.57.9 Osi Model: UGC NET CSE | June 2010 | Part 2 | Question: 29



What services does the internet layer provide ?

1. Quality of service
2. Routing
3. Addressing
4. Connection oriented delivery
5. Framing bits

- A. 1,2,3
- B. 2,3,4
- C. 1,3,4,5
- D. 2,3,4,5

ugcnetcse-june2010-paper2 computer-networks osi-model

Answer key

5.57.10 Osi Model: UGC NET CSE | June 2013 | Part 3 | Question: 18



Match the following :

- | | |
|----------------------|---------------------------|
| a. Data link Layer | i. Flow Control |
| b. Network Layer | ii. Node to node delivery |
| c. Transport Layer | iii. Mail services |
| d. Application Layer | iv. Routing |

Codes :

- A. a-ii, b-i, c-iv, d-iii
- B. a-ii, b-iv, c-i, d-iii
- C. a-ii, b-i, c-iii, d-iv
- D. a-ii, b-iv, c-iii, d-i

ugcnetcse-june2013-paper3 computer-networks osi-model

Answer key

5.57.11 Osi Model: UGC NET CSE | June 2014 | Part 3 | Question: 70



Which layers of the OSI reference model are host-to-host layers ?

- A. Transport, Session, Presentation, Application
- B. Network, Transport, Session, Presentation
- C. Data-link, Network, Transport, Session
- D. Physical, Data-link, Network, Transport

ugcnetjune2014iii computer-networks osi-model

Answer key 

5.58

Osi Protocol (5)

5.58.1 Osi Protocol: UGC NET CSE | December 2012 | Part 3 | Question: 10



Match the following :

List-I

List-II

- | | |
|----------------------|-----------|
| a. Application layer | i. TCP |
| b. Transport Layer | ii. HDLC |
| c. Network Layer | iii. HTTP |
| d. Data Link Layer | iv. BGP |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-ii, b-i, c-iv, d-iii | B. a-iii, b-iv, c-i, d-ii |
| C. a-iii, b-i, c-iv, d-ii | D. a-ii, b-iv, c-i, d-iii |

ugcnetcse-dec2012-paper3 computer-networks osi-protocol

Answer key 

5.58.2 Osi Protocol: UGC NET CSE | January 2017 | Part 2 | Question: 26



Match the following Layers and Protocols for a user browsing with SSL :

- | | |
|-------------------------|----------|
| a. Application of layer | i. TCP |
| b. Transport layer | ii. IP |
| c. Network layer | iii. PPP |
| d. Datalink layer | iv. HTTP |

Code :

- | | |
|---------------------------|---------------------------|
| A. a-iv, b-i, c-ii, d-iii | B. a-iii, b-ii, c-i, d-iv |
| C. a-ii, b-iii, c-iv, d-i | D. a-iii, b-i, c-iv, d-ii |

ugcnetjan2017ii computer-networks osi-protocol

Answer key 

5.58.3 Osi Protocol: UGC NET CSE | June 2012 | Part 3 | Question: 11



X.25 is _____ Network

- | | |
|--|--|
| A. Connection Oriented Network | B. Connection Less Network |
| C. Either Connection Oriented or Connection Less | D. Neither Connection Oriented nor Connection Less |

ugcnetcse-june2012-paper3 computer-networks osi-protocol

Answer key 

5.58.4 Osi Protocol: UGC NET CSE | June 2012 | Part 3 | Question: 41



Which layer of OSI reference model uses the ICMP (Internet Control Message Protocol)?

- | | |
|--------------------|----------------------|
| A. Transport layer | B. Data link layer |
| C. Network layer | D. Application layer |

ugcnetcse-june2012-paper3 computer-networks osi-protocol

Answer key 

5.58.5 Osi Protocol: UGC NET CSE | Junet 2015 | Part 2 | Question: 26



Which of the following protocols is an application layer protocol that establishes, manages and terminates multimedia sessions?

- A. Session Maintenance Protocol
- B. Real-time Streaming Protocol
- C. Real-time Transport Control Protocol
- D. Session Initiation Protocol

ugcnetcse-june2015-paper2 osi-protocol computer-networks

Answer key

5.59

Physical Layer (4)

5.59.1 Physical Layer: UGC NET CSE | December 2010 | Part 2 | Question: 28



How much bandwidth is there in 1 micron of spectrum at a wavelength of 1 micron ?

- A. 300 MHz
- B. 3 GHz
- C. 300 THz
- D. 30 KHz

ugcnetcse-dec2010-paper2 computer-networks physical-layer

Answer key

5.59.2 Physical Layer: UGC NET CSE | December 2013 | Part 2 | Question: 3



An analog signal carries 4 bits in each signal unit. If 1000 signal units are sent per second, then baud rate and bit rate of the signal are _____ and _____

- A. 4000 bauds/sec & 1000 bps
- B. 2000 bauds/sec & 1000 bps
- C. 1000 bauds/sec & 500 bps
- D. 1000 bauds/sec & 4000 bps

ugcnetcse-dec2013-paper2 computer-networks osi-model physical-layer

Answer key

5.59.3 Physical Layer: UGC NET CSE | July 2018 | Part 2 | Question: 42



The period of a signal is 100 ms. Its frequency is _____

- A. 100^3 Hertz
- B. 10^{-2} KHz
- C. 10^{-3} KHz
- D. 10^5 Hertz

ugcnetcse-july2018-paper2 computer-networks physical-layer

Answer key

5.59.4 Physical Layer: UGC NET CSE | June 2013 | Part 3 | Question: 21



Which one of the following media is multidrop?

- A. Shielded Twisted pair cable
- B. Unshielded Twisted pair cable
- C. Thick Coaxial cable
- D. Fiber Optic cable

ugcnetcse-june2013-paper3 computer-networks physical-layer

Answer key

5.60

Port Numbers (1)

5.60.1 Port Numbers: UGC NET CSE | Junet 2015 | Part 2 | Question: 27



Match the following port numbers with their uses:

List – I	List – II
a. 23	i. World wide web
b. 25	ii. Remote login
c. 80	iii. USENET news
d. 119	iv. E-mail

- A. $a - iv, b - i, c - ii, d - iii$
 C. $a - ii, b - iv, c - iii, d - i$

- B. $a - ii, b - i, c - iv, d - iii$
 D. $a - ii, b - iv, c - i, d - iii$

ugcnetcse-june2015-paper2 computer-networks port-numbers match-the-following

Answer key 

5.61

Propagation Delay (1)

5.61.1 Propagation Delay: UGC NET CSE | June 2019 | Part 2 | Question: 83



You are designing a link layer protocol for a link with bandwidth of 1 Gbps (10^9 bits/second) over a fiber link with length of 800 km. Assume the speed of light in this medium is 200000 km/second. What is the propagation delay in this link?

- A. 1 millisecond
 C. 3 milliseconds
 B. 2 milliseconds
 D. 4 milliseconds

ugcnetcse-june2019-paper2 propagation-delay

Answer key 

5.62

Propagation Time (1)

5.62.1 Propagation Time: UGC NET CSE | December 2014 | Part 3 | Question: 28



What is the propagation time if the distance between the two points is 48,000 ? Assume the propagation speed to be 2.4×10^8 metre/second in cable.

- A. 0.5 ms
 B. 20 ms
 C. 50 ms
 D. 200 ms

ugcnetcse-dec2014-paper3 computer-networks propagation-time

Answer key 

5.63

Protocol (1)

5.63.1 Protocol: UGC NET CSE | December 2014 | Part 2 | Question: 50



Match the following :

List – I

- a. Call Control protocol
 b. A-bis
 c. BSMAP
 d. CDMA

List – II

- i. Interface between Base Transceiver Station (BTS) and Base Station Controller (BSC)
 ii. Spread spectrum
 iii. Connection management
 iv. Works between Mobile Switching Centre (MSC) and Base Station Subsystem (BSS)

Codes :

- A. a-iii, b-iv, c-i, d-ii
 C. a-i, b-ii, c-iii, d-iv
 B. a-iii, b-i, c-iv, d-ii
 D. a-iv, b-iii, c-ii, d-i

ugcnetcse-dec2014-paper2 computer-networks protocol

Answer key 

5.64

Rarp Protocol (1)

5.64.1 Rarp Protocol: UGC NET CSE | June 2019 | Part 2 | Question: 82



What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?

- A. ARP
 B. DNS
 C. RARP
 D. ICMP

ugcnetcse-june2019-paper2 rarp-protocol

Answer key

5.65

Routers Bridge Hubs Switches (2)

5.65.1 Routers Bridge Hubs Switches: UGC NET CSE | June 2009 | Part 2 | Question: 49



The device which connects dissimilar *LANs* of different topologies using different sets of communication protocols so that information can flow from one to another is called :

- A. Router
- B. Bridge
- C. Gateway
- D. Switch

ugcnetcse-june2009-paper2 computer-networks routers-bridge-hubs-switches

Answer key

5.65.2 Routers Bridge Hubs Switches: UGC NET CSE | June 2014 | Part 2 | Question: 02



A bridge has access to _____ address in the same network.

- | | |
|-------------|----------------|
| A. Physical | B. Network |
| C. Datalink | D. Application |

ugcnetcse-june2014-paper2 routers-bridge-hubs-switches

Answer key

5.66

Routing (4)

5.66.1 Routing: UGC NET CSE | December 2011 | Part 2 | Question: 12



A station in a network in a network forward incoming packets by placing them on its shortest output queue. What routing algorithm is being used ?

- | | |
|-----------------------|------------------|
| A. Hot potato routing | B. Flooding |
| C. Static routing | D. Delta routing |

ugcnetcse-dec2011-paper2 computer-networks routing

Answer key

5.66.2 Routing: UGC NET CSE | December 2015 | Part 3 | Question: 12



Consider a subnet with 720 routers. If a three level hierarchy is chosen with 8 clusters, each containing 9 regions of 10 routers, then total number of entries in the routing table is _____

- | | | | |
|-------|-------|-------|-------|
| A. 25 | B. 27 | C. 53 | D. 72 |
|-------|-------|-------|-------|

computer-networks routing ugcnetcse-dec2015-paper3

Answer key

5.66.3 Routing: UGC NET CSE | June 2012 | Part 2 | Question: 17



In which routing method do all the routers have a common database?

- | | |
|--------------------|--------------------|
| A. Distance vector | B. Link state |
| C. Link vector | D. Dijkstra method |

ugcnetcse-june2012-paper2 computer-networks routing

Answer key

5.66.4 Routing: UGC NET CSE | June 2012 | Part 2 | Question: 41



What is the routing algorithm used by RIP and IGRP?

A. OSPF

B. Link-state

C. Dynamic

D. Dijkstra vector

computer-networks routing ugcnetcse-june2012-paper2

Answer key 

5.67

Routing Algorithm (3)

5.67.1 Routing Algorithm: UGC NET CSE | December 2008 | Part 2 | Question: 19



An example of a non-adaptive routing algorithm is:

- A. Shortest path routing
- B. Centralised routing
- C. Baran's hot potato routing
- D. Baran's backward learning algorithm

ugcnetcse-dec2008-paper2 computer-networks routing-algorithm

Answer key 

5.67.2 Routing Algorithm: UGC NET CSE | December 2014 | Part 2 | Question: 28



Which of the following algorithms is not a broadcast routing algorithm ?

- A. Flooding
- B. Multidestination routing
- C. Reverse path forwarding
- D. All of the above

ugcnetcse-dec2014-paper2 computer-networks routing-algorithm

Answer key 

5.67.3 Routing Algorithm: UGC NET CSE | September 2013 | Part 2 | Question: 6



The count-to-infinity problem is associated with

- A. Flooding algorithm
- B. Hierarchical routing algorithm
- C. Distance vector routing algorithm
- D. Link state routing algorithm

ugcnetsep2013ii computer-networks routing-algorithm

Answer key 

5.68

Rsa (1)

5.68.1 Rsa: UGC NET CSE | June 2013 | Part 2 | Question: 33



Using data $p=3$, $q=11$, $n=pq$, $d=7$ in RSA algorithm find the cipher text of the given plain text SUZANNE

- A. BUTAEEZ
- B. SUZANNE
- C. XYZABCD
- D. ABCDXYZ

computer-networks network-security rsa ugcnetcse-june2013-paper2

Answer key 

5.69

Rsa Public Key (1)

5.69.1 Rsa Public Key: UGC NET CSE | December 2013 | Part 2 | Question: 5



Using the RSA public key crypto system, if $p=13$, $q=31$ and $d=7$, then the value of e is

- A. 101
- B. 103
- C. 105
- D. 107

ugcnetcse-dec2013-paper2 computer-networks network-security rsa-public-key

Answer key 

5.70

Rsa Security Networks (2)

5.70.1 Rsa Security Networks: UGC NET CSE | January 2017 | Part 3 | Question: 29



In RSA public key cryptosystem suppose $n = p * q$ where p and q are primes. (e, n) and (d, n) are public and private keys respectively. Let M be an integer such that $0 < M < n$ and $\phi(n) = (p - 1)(q - 1)$.

Which of the following equations represent RSA public key cryptosystem?

- I.
 $C \equiv M^e \pmod{n}$
 $M \equiv C^d \pmod{n}$
 - II.
 $ed \equiv 1 \pmod{n}$
 - III.
 $ed \equiv 1 \pmod{\phi(n)}$
 - IV.
 $C \equiv M^e \pmod{\phi(n)}$
 $M \equiv C^d \pmod{\phi(n)}$
- A. I and II
B. I and III
C. II and III
D. I and IV

ugcnetcse-jan2017-paper3 computer-networks rsa-security-networks

Answer key

5.70.2 Rsa Security Networks: UGC NET CSE | June 2019 | Part 2 | Question: 90



The RSA encryption algorithm also works in reverse, that is, you can encrypt a message with the private key and decrypt it using the public key. This property is used in

- A. instruction detection systems
B. digital signatures
C. data compression
D. certification

ugcnetcse-june2019-paper2 network-security rsa-security-networks computer-networks

Answer key

5.71 Session Layer (1)

5.71.1 Session Layer: UGC NET CSE | Junet 2015 | Part 2 | Question: 28



Which of the following is not associated with the session layer?

- A. Dialog control
B. Token management
C. Semantics of the information transmitted
D. synchronization

ugcnetcse-june2015-paper2 computer-networks session-layer

Answer key

5.72 Signalling (1)

5.72.1 Signalling: UGC NET CSE | August 2016 | Part 3 | Question: 59



Which of the following is a source coding technique ?

- A. Huffman coding
B. Arithmetic coding
C. Run-length coding
D. DPCM

ugcnetcse-aug2016-paper3 computer-networks signalling

Answer key

5.73 Signals (1)

5.73.1 Signals: UGC NET CSE | June 2013 | Part 3 | Question: 20



The _____ measure the relative strengths of two signals or a signal at two different points.

- A. frequency B. attenuation C. throughput D. decibel

ugcnetcse-june2013-paper3 computer-networks signals

Answer key 

5.74 Sliding Window (4)

5.74.1 Sliding Window: UGC NET CSE | December 2009 | Part 2 | Question: 27



What is the maximum window size in sliding window protocol used in a computer network ?

- (A) 4
(B) 8
(C) 15
(D) 16

ugcnetcse-dec2009-paper2 computer-networks sliding-window

Answer key 

5.74.2 Sliding Window: UGC NET CSE | December 2012 | Part 3 | Question: 4



The technique of temporarily delaying outgoing acknowledgements so that they can be hooked onto the next outgoing data frame is known as

- A. Bit stuffing B. Piggy backing C. Pipelining D. Broadcasting

ugcnetcse-dec2012-paper3 computer-networks sliding-window

Answer key 

5.74.3 Sliding Window: UGC NET CSE | January 2017 | Part 3 | Question: 26



Station A uses 32 byte packets to transmit messages to station B using sliding window protocol. The round trip delay between A and B is 40 milliseconds and the bottleneck bandwidth on the path between A and B is 64 kbps. The optimal window size of A is

- A. 20 B. 10 C. 30 D. 40

ugcnetcse-jan2017-paper3 computer-networks sliding-window

Answer key 

5.74.4 Sliding Window: UGC NET CSE | June 2014 | Part 3 | Question: 72



Four bits are used for packed sequence numbering in a sliding window protocol used in a computer network. What is the maximum window size?

- A. 4 B. 8 C. 15 D. 16

ugcnetjune2014iii computer-networks sliding-window

Answer key 

5.75 Slotted Aloha (1)

5.75.1 Slotted Aloha: UGC NET CSE | July 2018 | Part 2 | Question: 41



A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200 Kbps bandwidth. Find the throughput of the system (all stations put together) produces 250 frames per second:

- A. 49 B. 368 C. 149 D. 151

Answer key 

5.76

Subnetting (2)

5.76.1 Subnetting: UGC NET CSE | December 2018 | Part 2 | Question: 87



An Internet Service Provider (*ISP*) has following chunk of *CIDR*-based *IP* addresses available with it : 245.248.128.0/20. The *ISP* wants to give half of this chunk addresses to organization *A* and a quarter to organization *B* while retaining the remaining with itself. Which of the following is a valid allocation of addresses to *A* and *B*?

- A. 245.248.136.0/21 and 245.248.128.0/22
 B. 245.248.128.0/21 and 245.248.128.0/22
 C. 245.248.132.0/22 and 245.248.132.0/21
 D. 245.248.136.0/24 and 245.248.132.0/21

ugcnetcse-dec2018-paper2 computer-networks subnetting

5.76.2 Subnetting: UGC NET CSE | June 2019 | Part 2 | Question: 84



You need 500 subnets, each with about 100 usable host address per subnet. What network mask will you assign using a class B network address?

- A. 255.255.255.252
 B. 255.255.255.128
 C. 255.255.255.0
 D. 255.255.254.0

ugcnetcse-june2019-paper2 subnetting

Answer key 

5.77

TCP (4)

5.77.1 TCP: UGC NET CSE | January 2017 | Part 2 | Question: 28



A packet whose destination is outside the local *TCP/IP* network segment is sent to _____.

- A. File server
 B. *DNS* server
 C. *DHCP* server
 D. Default gateway

ugcnetjan2017ii computer-networks tcp

Answer key 

5.77.2 TCP: UGC NET CSE | Junet 2015 | Part 3 | Question: 26



Which one of the following control fields in TCP header is used to specify whether the sender has no more data to transmit?

- A. FIN
 B. RST
 C. SYN
 D. PSH

ugcnetcse-june2015-paper3 computer-networks tcp

Answer key 

5.77.3 TCP: UGC NET CSE | Junet 2015 | Part 3 | Question: 30



The minimum payload of a TCP segment is

- A. 65,535
 B. 65,515
 C. 65,495
 D. 65,475

ugcnetcse-june2015-paper3 computer-networks tcp

Answer key 

5.77.4 TCP: UGC NET CSE | September 2013 | Part 3 | Question: 9



Which of the following field of the TCP header tells how many bytes may be sent starting at the byte acknowledged?

- A. TCP header length
 B. Window size
 C. Acknowledgement number
 D. Urgent pointer

Answer key **5.78 TCP Header (1)****5.78.1 TCP Header: UGC NET CSE | December 2014 | Part 3 | Question: 27**

Which of the following is not a field in *TCP* header ?

- A. Sequence number B. Fragment offset
C. Checksum D. Window size

ugcnetcse-dec2014-paper3 computer-networks tcp-header

Answer key **5.79 TCP IP Model (1)****5.79.1 TCP IP Model: UGC NET CSE | July 2016 | Part 2 | Question: 28**

In TCP/IP reference model, the job of _____ layer is to permit hosts to inject packets into any network and travel them independently to the destination

- A. Physical B. Transport C. Application D. Host-to-network

ugcnetcse-july2016-paper2 computer-networks tcp-ip-model

Answer key **5.80 TCP IP Suit Protocols (1)****5.80.1 TCP IP Suit Protocols: UGC NET CSE | December 2012 | Part 3 | Question: 22**

Which provides an interface to the TCP/IP suit protocols in Windows 95 and Windows NT?

- A. FTP Active-X Control B. TCP/IP Active-X Control
C. Calinsock Active-X Control D. HTML Active-X Control

ugcnetcse-dec2012-paper3 computer-networks tcp-ip-suit-protocols

Answer key **5.81 Tele Communication (1)****5.81.1 Tele Communication: UGC NET CSE | Junet 2015 | Part 3 | Question: 25**

Which transmission technique guarantees that data packets will be received by the receiver in the same order in which they were sent by the sender?

- A. Broadcasting B. Unicasting
C. Packet switching D. Circuit switching

ugcnetcse-june2015-paper3 computer-networks data-communication tele-communication

Answer key **5.82 Token Bucket (1)****5.82.1 Token Bucket: UGC NET CSE | January 2017 | Part 3 | Question: 30**

A node *X* on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. Token bucket is initially filled with 16 megabits. The maximum duration taken by *X* to transmit at full rate of 10 Mbps is _____secs.

- A. 1 B. 2 C. 3 D. 4

ugcnetcse-jan2017-paper3 computer-networks token-bucket

Answer key 

5.83.1 Transmission Media: UGC NET CSE | August 2016 | Part 2 | Question: 26

Which of the following statements is not true with respect to microwaves ?

- A. Electromagnetic waves with frequencies from 300 GHz to 400 THz.
- B. Propagation is line-of-sight.
- C. Very high-frequency waves cannot penetrate walls.
- D. Use of certain portions of the band requires permission from authorities.

ugcnetcse-aug2016-paper2 computer-networks transmission-media

Answer key

5.83.2 Transmission Media: UGC NET CSE | August 2016 | Part 2 | Question: 27

In a fast Ethernet cabling, 100 Base-*TX* uses _____ cable and maximum segment size is _____.

- A. Twisted pair, 100 metres
- B. Twisted pair, 200 metres
- C. Fibre optics, 1000 metres
- D. Fibre optics, 2000 metres

ugcnetcse-aug2016-paper2 computer-networks transmission-media

Answer key

5.83.3 Transmission Media: UGC NET CSE | August 2016 | Part 2 | Question: 28

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network ?

- A. 1 Mbps
- B. 2 Mbps
- C. 10 Mbps
- D. 12 Mbps

ugcnetcse-aug2016-paper2 computer-networks transmission-media

Answer key

5.83.4 Transmission Media: UGC NET CSE | August 2016 | Part 3 | Question: 29

An analog signal has a bit rate of 8000 bps and a baud rate of 1000. Then analog signal has _____ signal element and carry _____ data elements in each signal.

- A. 256,8 bits
- B. 128,4 bits
- C. 256,4 bits
- D. 128,8 bits

ugcnetcse-aug2016-paper3 computer-networks transmission-media

Answer key

5.83.5 Transmission Media: UGC NET CSE | December 2008 | Part 2 | Question: 10

Transmission of N signals, each band limited to f_m Hz by TDM, requires a minimum band-width of:

- A. f_m
- B. $2 f_m$
- C. $N f_m$
- D. $2N f_m$

ugcnetcse-dec2008-paper2 computer-networks transmission-media

Answer key

5.83.6 Transmission Media: UGC NET CSE | December 2008 | Part 2 | Question: 7

The *ATM* cell is:

1. 48 bytes long

2. 53 bytes long
3. 64 bytes long
4. 69 bytes long

ugcnetcse-dec2008-paper2 computer-networks transmission-media

Answer key 

5.83.7 Transmission Media: UGC NET CSE | December 2010 | Part 2 | Question: 26



Frequency shift keying is used mostly in

- | | |
|-----------------------|----------------------|
| A. Radio transmission | B. Telegraphy |
| C. Telephone | D. None of the above |

ugcnetcse-dec2010-paper2 computer-networks transmission-media

Answer key 

5.83.8 Transmission Media: UGC NET CSE | December 2012 | Part 2 | Question: 23



Coaxial cables are categorized by Radio Government rating are adapted for specialized functions. Category RG-59 with impedance 75Ω used for

- | | | | |
|-------------|-------------|------------------|-------------------|
| A. Cable TV | B. Ethernet | C. Thin Ethernet | D. Thick Ethernet |
|-------------|-------------|------------------|-------------------|

ugcnetcse-dec2012-paper2 computer-networks transmission-media

Answer key 

5.83.9 Transmission Media: UGC NET CSE | July 2016 | Part 2 | Question: 30



Optical fiber uses reflection to guide light through a channel, in which angle of incidence is ____ the critical angle

- | | |
|-----------------|--------------------------|
| A. equal to | B. less than |
| C. greater than | D. less than or equal to |

ugcnetcse-july2016-paper2 computer-networks transmission-media

Answer key 

5.84 Unguided Media (1)

5.84.1 Unguided Media: UGC NET CSE | December 2013 | Part 2 | Question: 4



The VLF and LF bauds use ____ propagation for communication

- | | | | |
|-----------|--------|------------------|----------|
| A. Ground | B. Sky | C. Line of sight | D. Space |
|-----------|--------|------------------|----------|

ugcnetcse-dec2013-paper2 computer-networks unguided-media

Answer key 

5.85 Wifi (1)

5.85.1 Wifi: UGC NET CSE | December 2009 | Part 2 | Question: 49



Another name of IEEE 802.11 a is _____

- (A) Wi-Max
- (B) Fast Ethernet
- (C) Wi-Fi
- (D) 802.11 g

Answer key 

5.86

Wireless Communication (1)

5.86.1 Wireless Communication: UGC NET CSE | June 2012 | Part 2 | Question: 28



Mobile IP provides two basic functions

- | | |
|-------------------------------------|-------------------------------------|
| A. Route discovery and registration | B. Agent discovery and registration |
| C. IP binding and registration | D. None of the above |

ugcnetcse-june2012-paper2 computer-networks mobile-communication wireless-communication

Answer key 

5.87

Wireless Lan (4)

5.87.1 Wireless Lan: UGC NET CSE | December 2009 | Part 2 | Question: 48



Which technology is sometime referred to as wireless cable ?

- (A) MMDS
- (B) ATM
- (C) LMDS
- (D) CDMA

ugcnetcse-dec2009-paper2 computer-networks wireless-lan

Answer key 

5.87.2 Wireless Lan: UGC NET CSE | December 2009 | Part 2 | Question: 50



The unlicensed National Information Infrastructure band operates at the _____ frequency

- (A) 2.4 GHz
- (B) 5 GHz
- (C) 33 MHz
- (D) 5 MHz

ugcnetcse-dec2009-paper2 computer-networks wireless-lan

Answer key 

5.87.3 Wireless Lan: UGC NET CSE | December 2015 | Part 2 | Question: 21



Which of the following services is not provided by wireless access point in 802.11 WLAN?

- | | |
|---------------------|-------------------|
| A. Association | B. Disassociation |
| C. Error correction | D. Integration |

ugcnetcse-dec2015-paper2 computer-networks wireless-lan

Answer key 

5.87.4 Wireless Lan: UGC NET CSE | June 2010 | Part 2 | Question: 30



What is the maximum operating rate of a wireless LAN using infrared communication ?

- | | | | |
|-----------|-----------|-----------|------------|
| A. 1 mbps | B. 2 mbps | C. 5 mbps | D. 11 mbps |
|-----------|-----------|-----------|------------|

5.88

Wireless Networks (1)

5.88.1 Wireless Networks: UGC NET CSE | December 2008 | Part 2 | Question: 47



Assertion (A): Cellular telephone systems can handle a multitude of users .

Reasoning (R): Cellular telephone systems permit extensive frequency reuse in a small local area .

- A. Both (A) and (R) are true and (R) is the correct explanation for (A)
 B. (B) Both (A) and (R) are true but (R) is not the correct explanation
 C. (A) is true but (R) is false
 D. (A) is false but (R) is true

ugcnetcse-dec2008-paper2 computer-networks wireless-networks

5.89

Wireless Protocol (1)

5.89.1 Wireless Protocol: UGC NET CSE | December 2013 | Part 2 | Question: 42



Match the following :

List – I

- a. Wireless Application Environment
 b. Wireless Transaction protocol
 c. Wireless Datagram Protocol
 d. Wireless

List – II

- i. HTTP
 ii. IP
 iii. Scripts
 iv. UDP

Codes :

- A. a-ii, b-iv, c-i, d-iii
 B. a-iv, b-iii, c-ii, d-i
 C. a-iv, b-iii, c-i, d-ii
 D. a-iii, b-i, c-iv, d-ii

ugcnetcse-dec2013-paper2 computer-networks wireless-protocol

Answer key

5.90

Wireless Transmission (1)

5.90.1 Wireless Transmission: UGC NET CSE | June 2012 | Part 3 | Question: 66



If the period of a signal is 1000 ms, then what is its frequency in kilohertz?

- A. 10^{-3} KHz B. 10^{-2} KHz C. 10^{-1} KHz D. 1 KHz

ugcnetcse-june2012-paper3 computer-networks data-communication wireless-transmission

Answer key

Answer Keys

5.0.1	B	5.0.2	D	5.0.3	A	5.0.4	D	5.0.5	X
5.0.6	D	5.0.7	B	5.0.8	C	5.0.9	B	5.0.10	B
5.0.11	B	5.0.12	D	5.0.13	C	5.0.14	B	5.0.15	A
5.0.16	D	5.0.17	B	5.0.18	C	5.0.19	C	5.0.20	D
5.0.21	B	5.0.22	Q-Q	5.0.23	Q-Q	5.0.24	D	5.0.25	Q-Q
5.0.26	Q-Q	5.0.27	Q-Q	5.0.28	B	5.0.29	D	5.0.30	Q-Q
5.0.31	Q-Q	5.0.32	Q-Q	5.0.33	Q-Q	5.0.34	Q-Q	5.0.35	Q-Q

5.0.36	Q-Q	5.0.37	Q-Q	5.0.38	Q-Q	5.0.39	Q-Q	5.0.40	Q-Q
5.0.41	D	5.0.42	Q-Q	5.0.43	Q-Q	5.0.44	Q-Q	5.0.45	Q-Q
5.0.46	Q-Q	5.0.47	Q-Q	5.0.48	Q-Q	5.0.49	Q-Q	5.0.50	Q-Q
5.0.51	Q-Q	5.0.52	Q-Q	5.0.53	Q-Q	5.0.54	Q-Q	5.0.55	Q-Q
5.0.56	Q-Q	5.0.57	Q-Q	5.0.58	Q-Q	5.0.59	A	5.0.60	A
5.0.61	Q-Q	5.1.1	D	5.2.1	B	5.3.1	X	5.4.1	A
5.5.1	C	5.6.1	Q-Q	5.7.1	A	5.8.1	B	5.9.1	B
5.10.1	Q-Q	5.11.1	D	5.11.2	A	5.12.1	Q-Q	5.13.1	Q-Q
5.14.1	A	5.14.2	X	5.15.1	D	5.16.1	C	5.16.2	D
5.16.3	A	5.16.4	B	5.16.5	C	5.16.6	D	5.16.7	A
5.16.8	C	5.16.9	C	5.17.1	D	5.18.1	Q-Q	5.18.2	C
5.18.3	D	5.18.4	Q-Q	5.18.5	C	5.18.6	C	5.18.7	C
5.19.1	Q-Q	5.19.2	Q-Q	5.19.3	Q-Q	5.19.4	Q-Q	5.19.5	Q-Q
5.19.6	Q-Q	5.19.7	C	5.20.1	Q-Q	5.20.2	Q-Q	5.20.3	B
5.20.4	D	5.20.5	D	5.20.6	D	5.20.7	C	5.21.1	Q-Q
5.22.1	B	5.23.1	Q-Q	5.23.2	C	5.24.1	Q-Q	5.25.1	B
5.26.1	D	5.26.2	A	5.27.1	C	5.27.2	D	5.28.1	Q-Q
5.29.1	Q-Q	5.29.2	C	5.29.3	Q-Q	5.29.4	Q-Q	5.29.5	C
5.30.1	X	5.31.1	A	5.31.2	B	5.31.3	D	5.31.4	B
5.31.5	A	5.32.1	A	5.33.1	D	5.34.1	D	5.35.1	Q-Q
5.35.2	Q-Q	5.35.3	Q-Q	5.35.4	C	5.35.5	Q-Q	5.35.6	Q-Q
5.35.7	D	5.35.8	C	5.35.9	A	5.35.10	C	5.35.11	A
5.35.12	B	5.35.13	A	5.36.1	A	5.36.2	D	5.37.1	B
5.38.1	Q-Q	5.39.1	A	5.40.1	D	5.41.1	Q-Q	5.41.2	A
5.42.1	B	5.43.1	B	5.44.1	D	5.44.2	C	5.44.3	D
5.45.1	A	5.45.2	C	5.45.3	B	5.45.4	D	5.45.5	A
5.45.6	B	5.45.7	B	5.46.1	C	5.46.2	C	5.47.1	C
5.48.1	Q-Q	5.49.1	Q-Q	5.49.2	Q-Q	5.50.1	B	5.50.2	D
5.51.1	A	5.52.1	A	5.52.2	C	5.52.3	C	5.52.4	A
5.52.5	A	5.53.1	Q-Q	5.53.2	Q-Q	5.53.3	Q-Q	5.53.4	C
5.53.5	Q-Q	5.53.6	C	5.53.7	D	5.53.8	B	5.53.9	D
5.53.10	C	5.54.1	Q-Q	5.54.2	Q-Q	5.54.3	B	5.54.4	C
5.54.5	D	5.54.6	X	5.54.7	Q-Q	5.54.8	Q-Q	5.54.9	C
5.54.10	B	5.54.11	A	5.54.12	Q-Q	5.54.13	Q-Q	5.54.14	A
5.54.15	D	5.54.16	B	5.54.17	C	5.54.18	A	5.55.1	Q-Q
5.55.2	Q-Q	5.55.3	D	5.55.4	Q-Q	5.55.5	B	5.56.1	Q-Q
5.56.2	Q-Q	5.56.3	A	5.56.4	A	5.57.1	Q-Q	5.57.2	Q-Q
5.57.3	Q-Q	5.57.4	Q-Q	5.57.5	A	5.57.6	B	5.57.7	C
5.57.8	Q-Q	5.57.9	Q-Q	5.57.10	B	5.57.11	A	5.58.1	C

5.58.2	A
5.59.2	D
5.62.1	X
5.66.1	Q-Q
5.67.2	X
5.70.2	B
5.74.2	B
5.76.2	B
5.78.1	B
5.83.1	Q-Q
5.83.6	Q-Q
5.85.1	Q-Q
5.87.4	Q-Q

5.58.3	A
5.59.3	Q-Q
5.63.1	B
5.66.2	A
5.67.3	C
5.71.1	C
5.74.3	B
5.77.1	D
5.79.1	X
5.83.2	Q-Q
5.83.7	Q-Q
5.86.1	B
5.88.1	Q-Q

5.58.4	C
5.59.4	C
5.64.1	C
5.66.3	B
5.68.1	A
5.72.1	Q-Q
5.74.4	C
5.77.2	A
5.80.1	X
5.83.3	Q-Q
5.83.8	A
5.87.1	Q-Q
5.89.1	D

5.58.5	D
5.60.1	D
5.65.1	C
5.66.4	D
5.69.1	B
5.73.1	D
5.75.1	Q-Q
5.77.3	D
5.81.1	D
5.83.4	Q-Q
5.83.9	C
5.87.2	Q-Q
5.90.1	A

5.59.1	Q-Q
5.61.1	D
5.65.2	A
5.67.1	Q-Q
5.70.1	B
5.74.1	Q-Q
5.76.1	Q-Q
5.77.4	B
5.82.1	B
5.83.5	Q-Q
5.84.1	A
5.87.3	C